



Partners from RF to Light

AMCAD Model v17

2 * 0.106 mm

Sullivan Plet & Christian Cassou –September 2017



- Pulsed IV
 - Thermal Validation
 - Id decreasing
 - Trap Validation
 - Transient versus Self Heating Ratio
- Pulsed S-Parameters
 - 28 V – 20 mA/mm
 - 50 V – 20 mA/mm
- Static S-Parameters
 - Class C & Class AB at 28 V
- LoadPull Performances
 - 1.8 GHz at MXP & MXE - 28 V
 - 3.8 GHz at MXP & MXE - 28 V

> Pulsed IV

- **Thermal Validation**
- Id decreasing
- Trap Validation
- Transient versus Self Heating Ratio

> Pulsed S-Parameters

- 28 V – 20 mA/mm
- 50 V – 20 mA/mm

> Static S-Parameters

- Class C & Class AB at 28 V

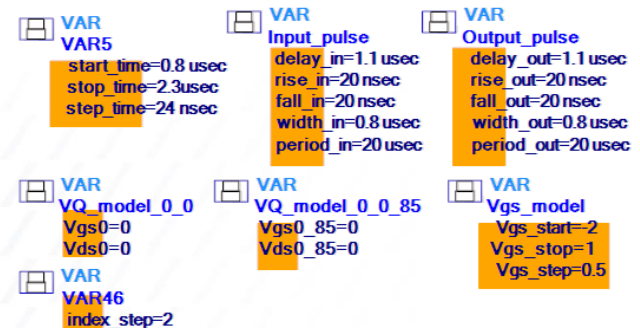
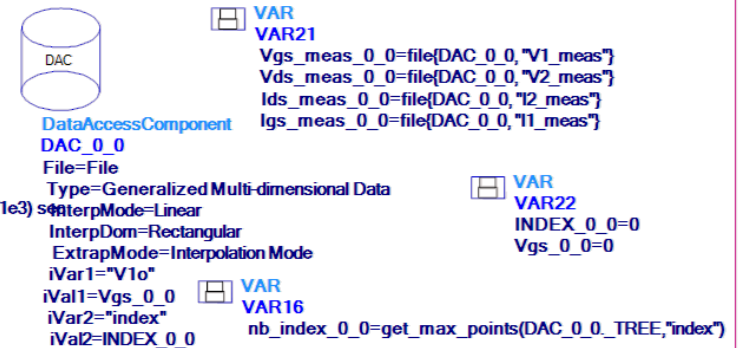
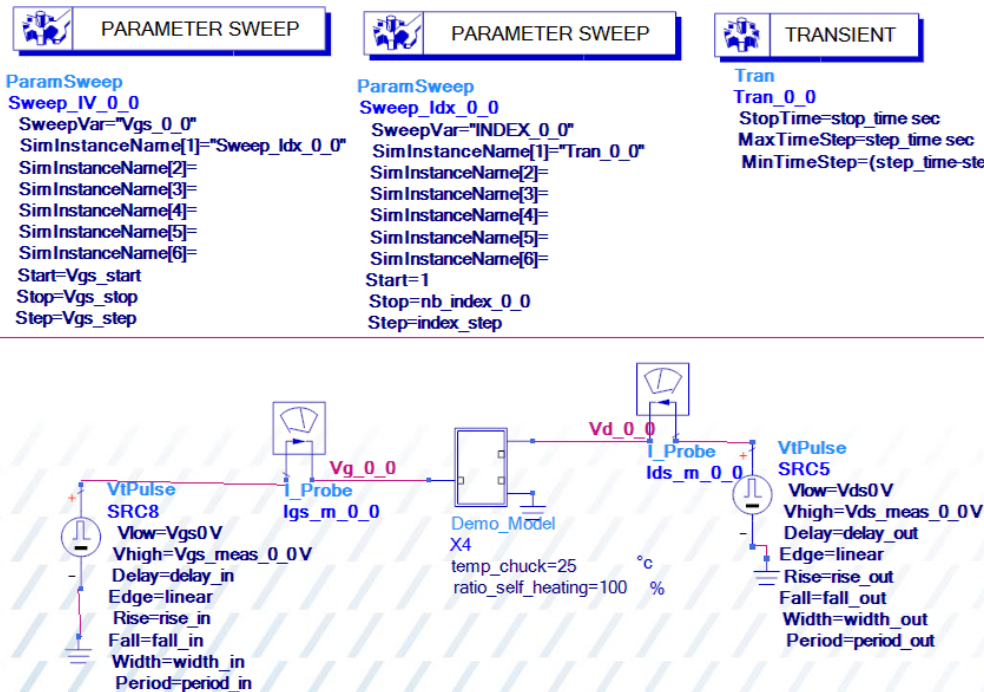
> LoadPull Performances

- 1.8 GHz at MXP & MXE - 28 V
- 3.8 GHz at MXP & MXE - 28 V

Transient Simulation – Thermal Validation

- T Chuck = 25° & Bias Point : $V_{gs0} = V_{ds0} = 0$ V
 - Transient simulation with self heating ratio = 100 %

I(V) Curves VQ(0,0) @ 25°C

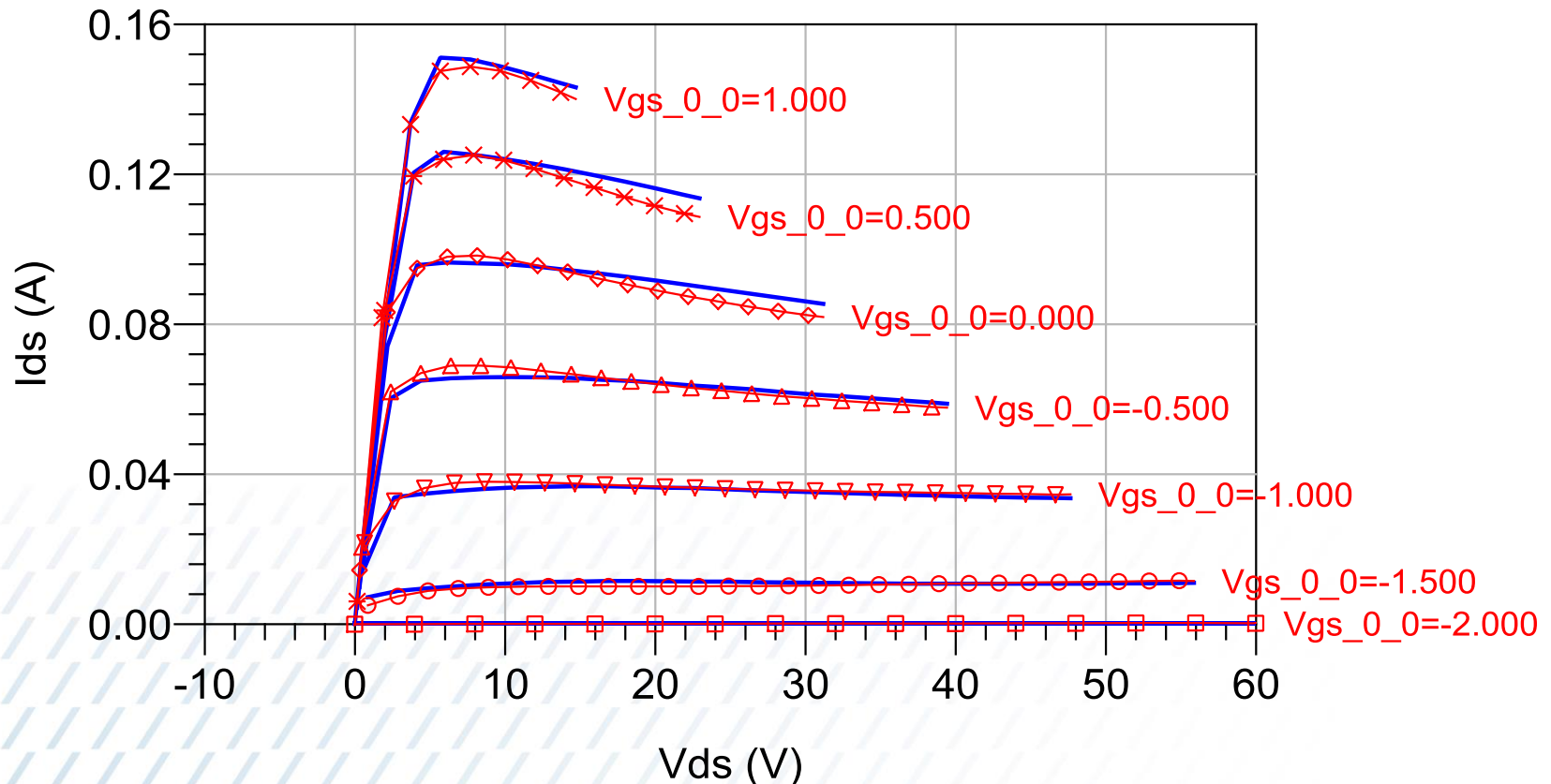


Pulsed IV

Transient Simulation – Thermal Validation

➤ T Chuck = 25° & Bias Point : $V_{gs0} = V_{ds0} = 0$ V

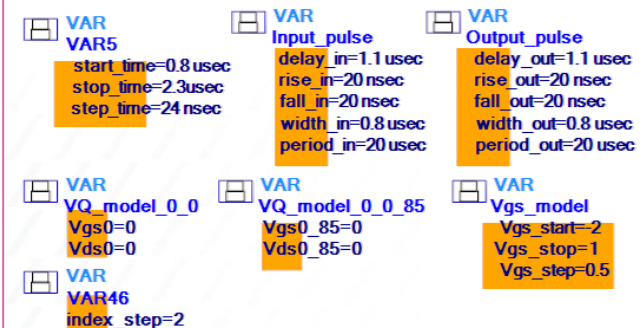
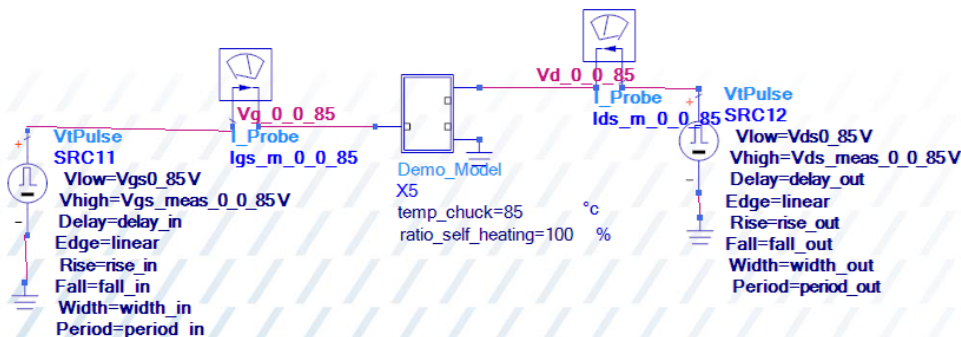
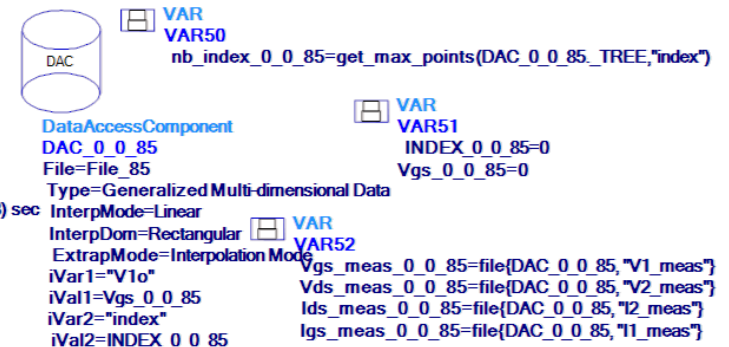
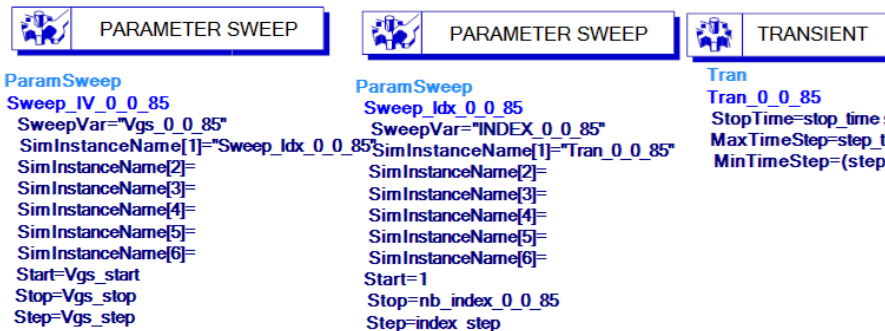
- Transient simulation with self heating ratio = 100 %



Transient Simulation – Thermal Validation

- T Chuck = 85° & Bias Point : $V_{gs0} = V_{ds0} = 0$ V
 - Transient simulation with self heating ratio = 100 %

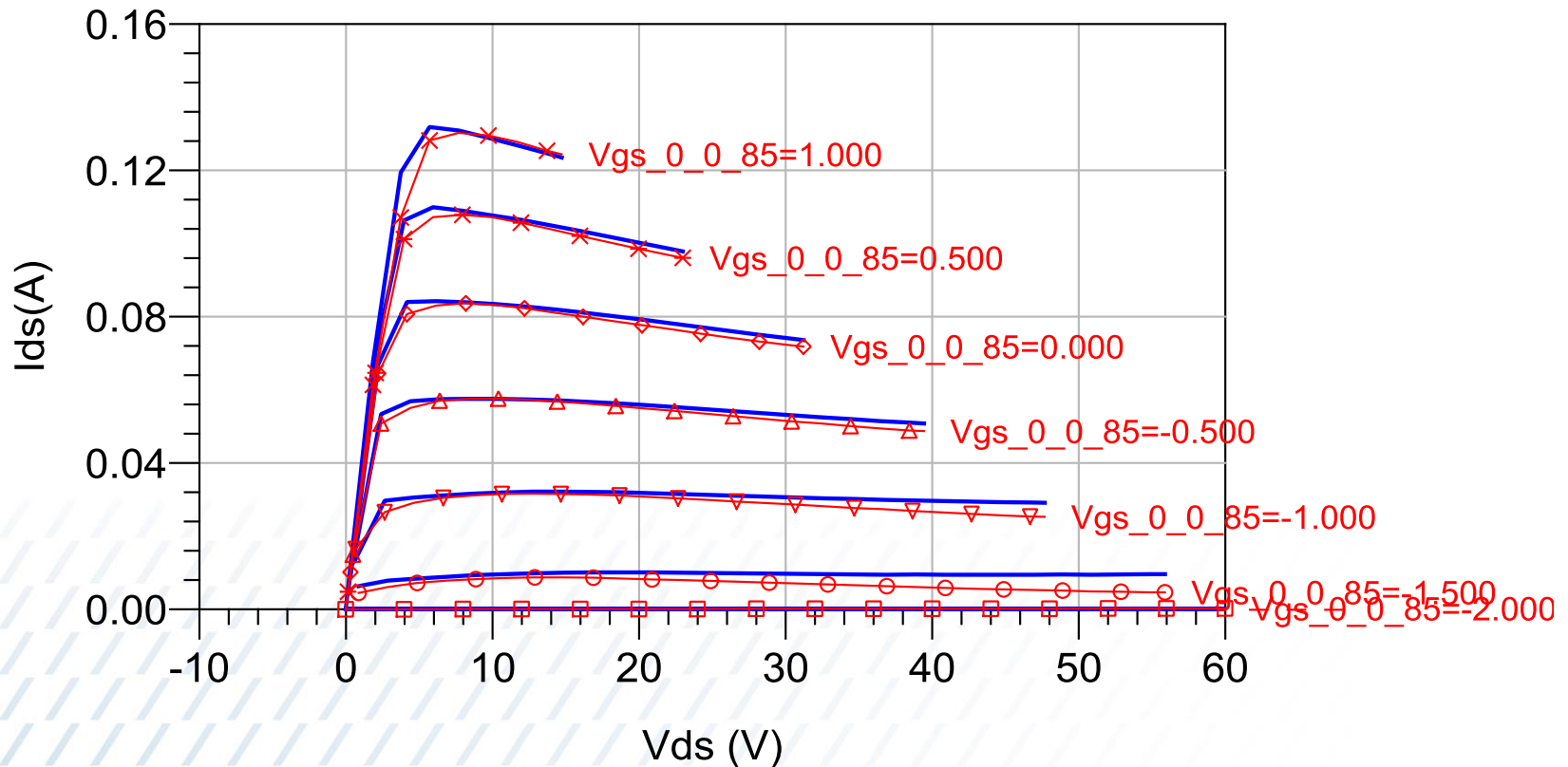
I(V) Curves VQ(0,0) @ 85°C



Transient Simulation – Thermal Validation

➤ T Chuck = 85° & Bias Point : $V_{gs0} = V_{ds0} = 0$ V

- Transient simulation with self heating ratio = 100 %



> Pulsed IV

- Thermal Validation
- **Id decreasing**
- Trap Validation
- Transient versus Self Heating Ratio

> Pulsed S-Parameters

- 28 V – 20 mA/mm
- 50 V – 20 mA/mm

> Static S-Parameters

- Class C & Class AB at 28 V

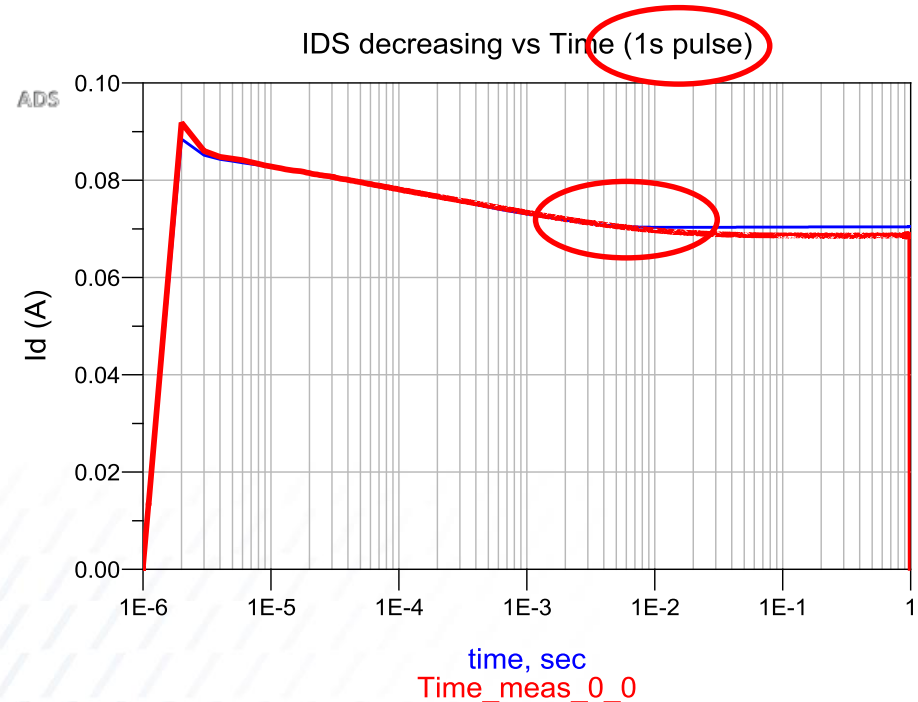
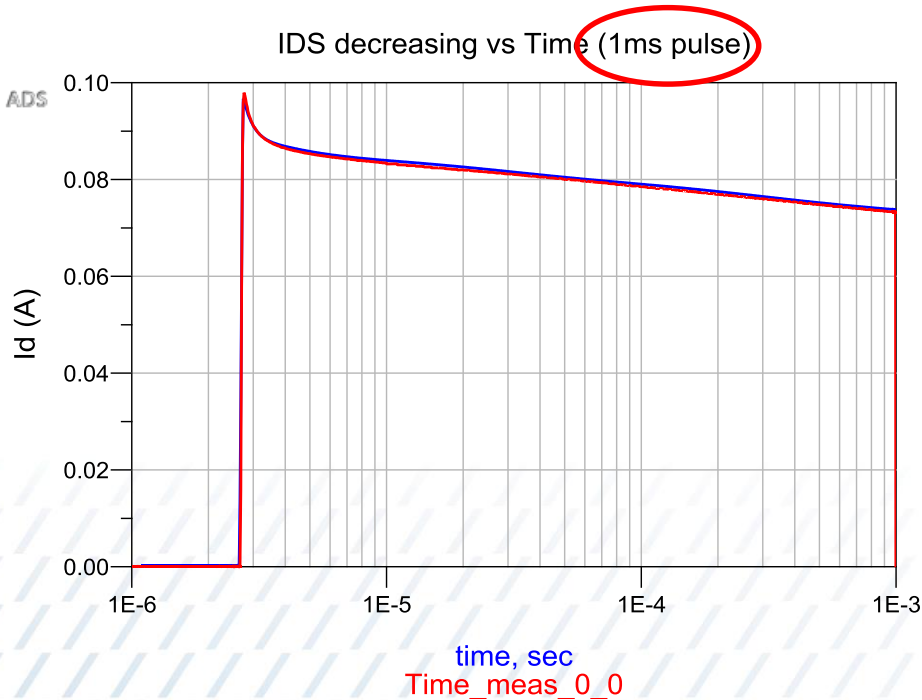
> LoadPull Performances

- 1.8 GHz at MXP & MXE - 28 V
- 3.8 GHz at MXP & MXE - 28 V

Id decreasing versus time

Transient Simulation with long pulse

- Thermal circuit is composed of 3 RC cells
 - model self-heating as a function of time



> Pulsed IV

- Thermal Validation
- Id decreasing
- **Trap Validation**
- Transient versus Self Heating Ratio

> Pulsed S-Parameters

- 28 V – 20 mA/mm
- 50 V – 20 mA/mm

> Static S-Parameters

- Class C & Class AB at 28 V

> LoadPull Performances

- 1.8 GHz at MXP & MXE - 28 V
- 3.8 GHz at MXP & MXE - 28 V

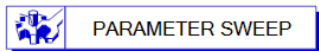
Trap Validation

Transient Simulation

➤ Gate Trap measurement at 500 ns

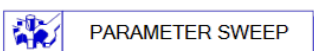
- $V_{gs0} = -2 \text{ V}$ & $V_{ds0} = 0 \text{ V}$

I(V) Curves VQ(-2,0) @ 25°C



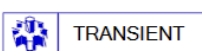
PARAMETER SWEEP

ParamSweep
Sweep_IV_2_0
SweepVar="Vgs_2_0"
SimInstanceName[1]="Sweep_idx_2_0"
SimInstanceName[2]=
SimInstanceName[3]=
SimInstanceName[4]=
SimInstanceName[5]=
SimInstanceName[6]=
Start=Vgs_start
Stop=Vgs_stop
Step=Vgs_step



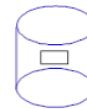
PARAMETER SWEEP

ParamSweep
Sweep_idx_2_0
SweepVar="INDEX_2_0"
SimInstanceName[1]="Tran_2_0"
SimInstanceName[2]=
SimInstanceName[3]=
SimInstanceName[4]=
SimInstanceName[5]=
SimInstanceName[6]=
Start=1
Stop=nb_index_2_0
Step=index_step



TRANSIENT

Tran
Tran_2_0
StopTime=stop_time sec
MaxTimeStep=step_time sec
MinTimeStep=(step_time-step_time/1e3) sec



DataAccessComponent
DAC_2_0
File=File
Type=Generalized Multi-dimensional Data
InterpMode=Linear
InterpDom=Rectangular
ExtrapMode=Interpolation Mode
iVal1="V1o"
iVal1=Vgs_2_0
iVal2="index"
iVal2=INDEX_2_0

VAR
VAR21

Vgs_meas_2_0=file{DAC_2_0,"V1_meas"}
Vds_meas_2_0=file{DAC_2_0,"V2_meas"}
Ids_meas_2_0=file{DAC_2_0,"I2_meas"}
Igs_meas_2_0=file{DAC_2_0,"I1_meas"}

VAR
VAR22

INDEX_2_0=0
Vgs_2_0=0

VAR
VAR16

nb_index_2_0=get_max_points(DAC_2_0_TREE,"index")

VAR
VAR56

start_time=0.8 usec
stop_time=2.3usec
step_time=24 nsec

VAR
Input_pulse1

delay_in=1.1 usec
rise_in=20 nsec
fall_in=20 nsec
width_in=0.8 usec
period_in=20 usec

VAR
Output_pulse1

delay_out=1.1 usec
rise_out=20 nsec
fall_out=20 nsec
width_out=0.8 usec
period_out=20 usec

VAR
File_2_0

File="IV_Pulsed_2x106um_sample_D1_Reverse_Vq_Vpm2_0_T20_iv.mdf"

VAR
VQ_model_2_0

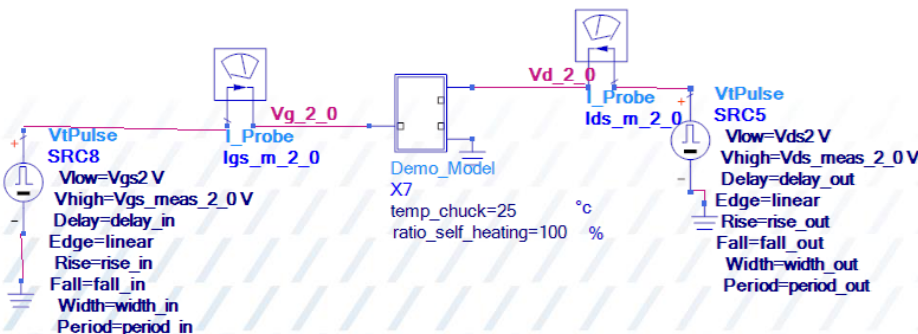
Vgs2=-2
Vds2=0

VAR
Vgs_model1

Vgs_start=-2
Vgs_stop=1
Vgs_step=0.5

VAR
VAR57

index_step=2

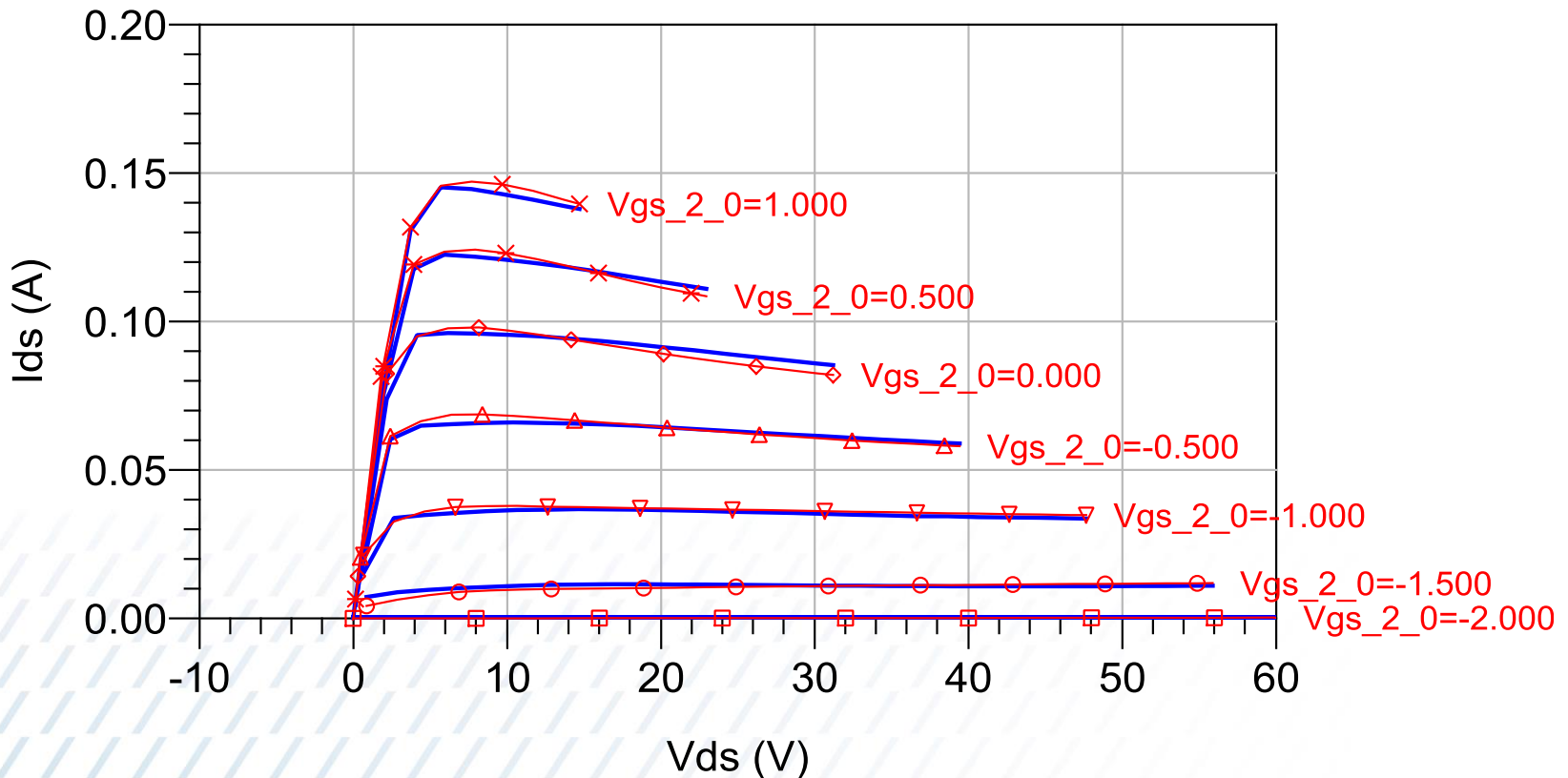


Trap Validation

Transient Simulation

➤ Gate Trap measurement at 500 ns

- $V_{gs0} = -2 \text{ V}$ & $V_{ds0} = 0 \text{ V}$



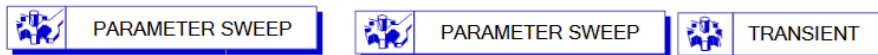
Trap Validation

Transient Simulation

➤ Gate & Drain Trap measurement at 500 ns

- $V_{gs0} = -2 \text{ V}$ & $V_{ds0} = 28 \text{ V}$

I(V) Curves VQ(-2,28) @ 25°C



ParamSweep

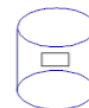
Sweep_IV_2_28
SweepVar="Vgs_2_28"
SimInstanceName[1]="Sweep_idx_2_28"
SimInstanceName[2]=
SimInstanceName[3]=
SimInstanceName[4]=
SimInstanceName[5]=
SimInstanceName[6]=
Start=Vgs_start
Stop=Vgs_stop
Step=Vgs_step

ParamSweep

Sweep_idx_2_28
SweepVar="INDEX_2_28"
SimInstanceName[1]="Tran_2_28"
SimInstanceName[2]=
SimInstanceName[3]=
SimInstanceName[4]=
SimInstanceName[5]=
SimInstanceName[6]=
Start=1
Stop=nb_index_2_28
Step=index_step

Tran

Tran_2_28
StopTime=stop_time sec
MaxTimeStep=step_time sec
MinTimeStep=(step_time-step_time/1e3) sec



VAR

VAR52
Vgs_meas_2_28=file[DAC_2_28,"V1_meas"]
Vds_meas_2_28=file[DAC_2_28,"V2_meas"]
Ids_meas_2_28=file[DAC_2_28,"I2_meas"]
Igs_meas_2_28=file[DAC_2_28,"I1_meas"]

DataAccessComponent

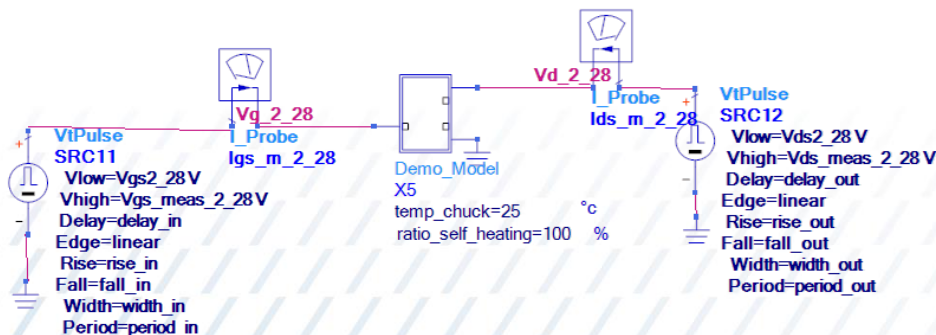
DAC_2_28
File=File_2_28
Type=Generalized Multi-dimensional Data
InterpMode=Linear
InterpDom=Rectangular
ExtrapMode=Interpolation Mode
iVar1="V1o"
iVal1=Vgs_2_28
iVar2="index"
iVal2=INDEX_2_28

VAR

VAR51
INDEX_2_28=0
Vgs_2_28=0

VAR

VAR50
nb_index_2_28=get_max_points(DAC_2_28_TREE,"index")



VAR

VAR5
start_time=0.8 usec
stop_time=2.3usec
step_time=24 nsec

VAR

Input_pulse
delay_in=1.1 usec
rise_in=20 nsec
fall_in=20 nsec
width_in=0.8 usec
period_in=20 usec

VAR

Output_pulse
delay_out=1.1 usec
rise_out=20 nsec
fall_out=20 nsec
width_out=0.8 usec
period_out=20 usec

VAR

File_2_28
File_2_28="IV_Pulsed_2x106um_sample_D1_Reverse_Vq_Vpm_2_28_T20_iv.mdf"

VAR

VQ_model_2_28
Vgs2_28=-2
Vds2_28=28

VAR

Vgs_model
Vgs_start=-2
Vgs_stop=1
Vgs_step=0.5

VAR

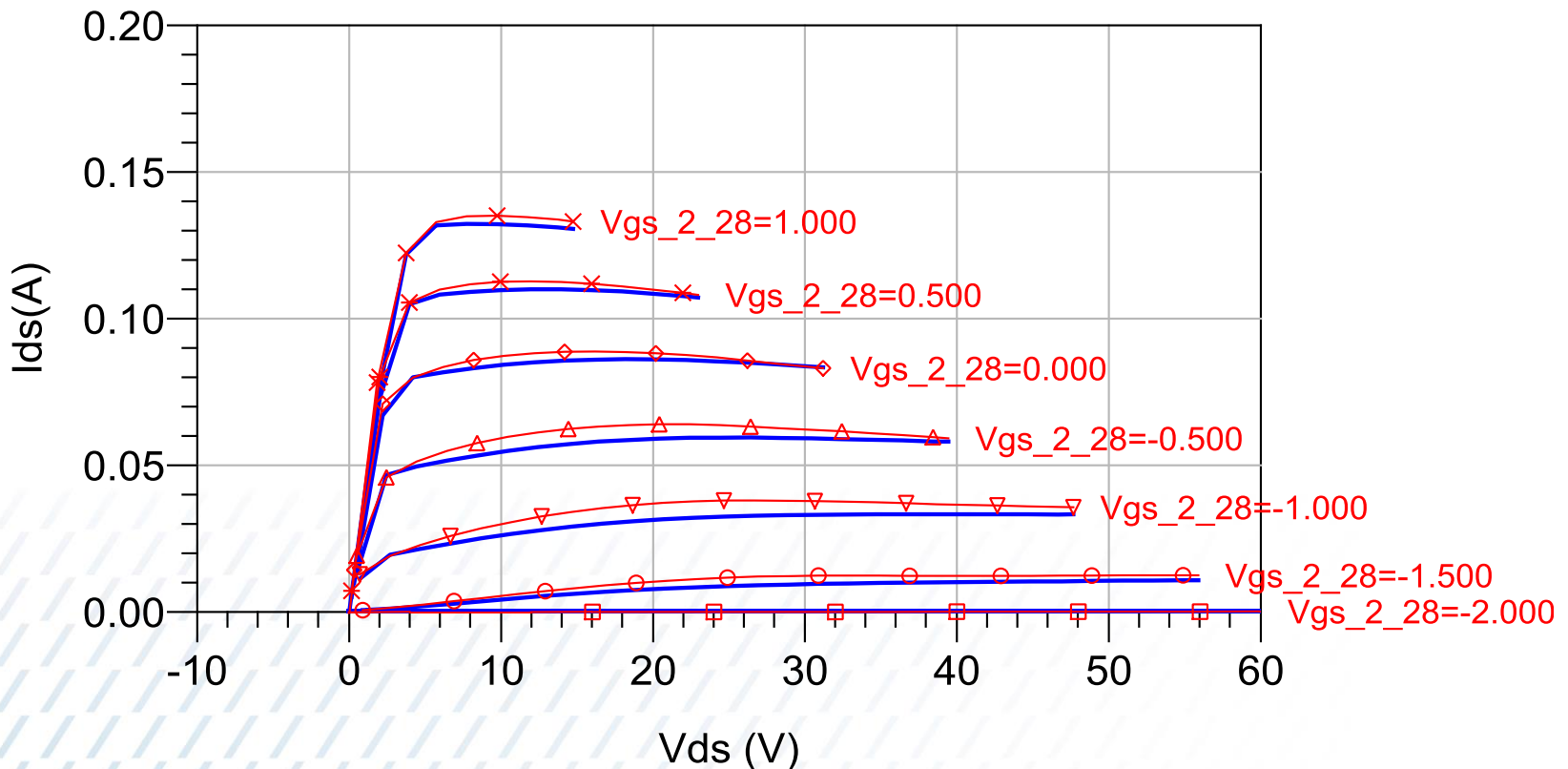
VAR46
index_step=2

Trap Validation

Transient Simulation

➤ Gate & Drain Trap measurement at 500 ns

- $V_{gs0} = -2 \text{ V}$ & $V_{ds0} = 28 \text{ V}$



Trap Validation

Transient Simulation

➤ Gate & Drain Trap measurement at 500 ns

- $V_{gs0} = -2 \text{ V}$ & $V_{ds0} = 50 \text{ V}$

I(V) Curves VQ(-2,50) @ 25°C

PARAMETER SWEEP

ParamSweep
Sweep_IV_2_50
SweepVar="Vgs_2_50"
SimInstanceName[1]="Sweep_idx_2_50"
SimInstanceName[2]=
SimInstanceName[3]=
SimInstanceName[4]=
SimInstanceName[5]=
SimInstanceName[6]=
Start=Vgs_start
Stop=Vgs_stop
Step=Vgs_step

PARAMETER SWEEP

ParamSweep
Sweep_idx_2_50
SweepVar="INDEX_2_50"
SimInstanceName[1]="Tran_2_50"
SimInstanceName[2]=
SimInstanceName[3]=
SimInstanceName[4]=
SimInstanceName[5]=
SimInstanceName[6]=
Start=1
Stop=nb_index_2_50
Step=index_step

TRANSIENT

Tran
Tran_2_50
StopTime=stop_time sec
MaxTimeStep=step_time sec
MinTimeStep=(step_time-step_time/1e3) sec



VAR
VAR53

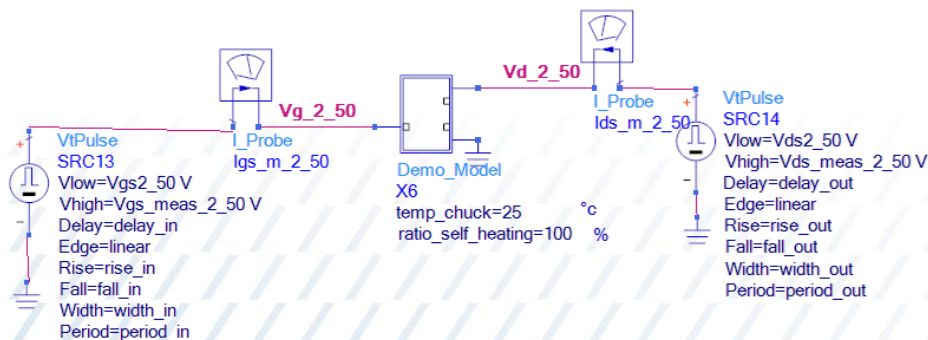
Vgs_meas_2_50=file[DAC_2_50,"V1_meas"]
Vds_meas_2_50=file[DAC_2_50,"V2_meas"]
Ids_meas_2_50=file[DAC_2_50,"I2_meas"]
Igs_meas_2_50=file[DAC_2_50,"I1_meas"]

DataAccessComponent
DAC_2_50
File=File_2_50
Type=Generalized Multi-dimensional Data
InterpMode=Linear
InterpDom=Rectangular
ExtrapMode=Interpolation Mode
iVar1="V1o"
iVal1=Vgs_2_50
iVar2="index"
iVal2=INDEX_2_50

VAR
VAR54
INDEX_2_50=0
Vgs_2_50=0

VAR
VAR55

nb_index_2_50=get_max_points(DAC_2_50_TREE,"index")



VAR
VAR5
start_time=0.8 usec
stop_time=2.3usec
step_time=24 nsec

VAR
Input_pulse
delay_in=1.1 usec
rise_in=20 nsec
fall_in=20 nsec
width_in=0.8 usec
period_in=20 usec

VAR
Output_pulse
delay_out=1.1 usec
rise_out=20 nsec
fall_out=20 nsec
width_out=0.8 usec
period_out=20 usec

VAR
File_2_50
File_2_50="IV_Pulsed_2x106µm_sample_D1_Reverse_Vq_Vpm2_50_T20_iv.mdf"

VAR
VQ_model_2_50
Vgs2_50=-2
Vds2_50=50

VAR
Vgs_model
Vgs_start=-2
Vgs_stop=1
Vgs_step=0.5

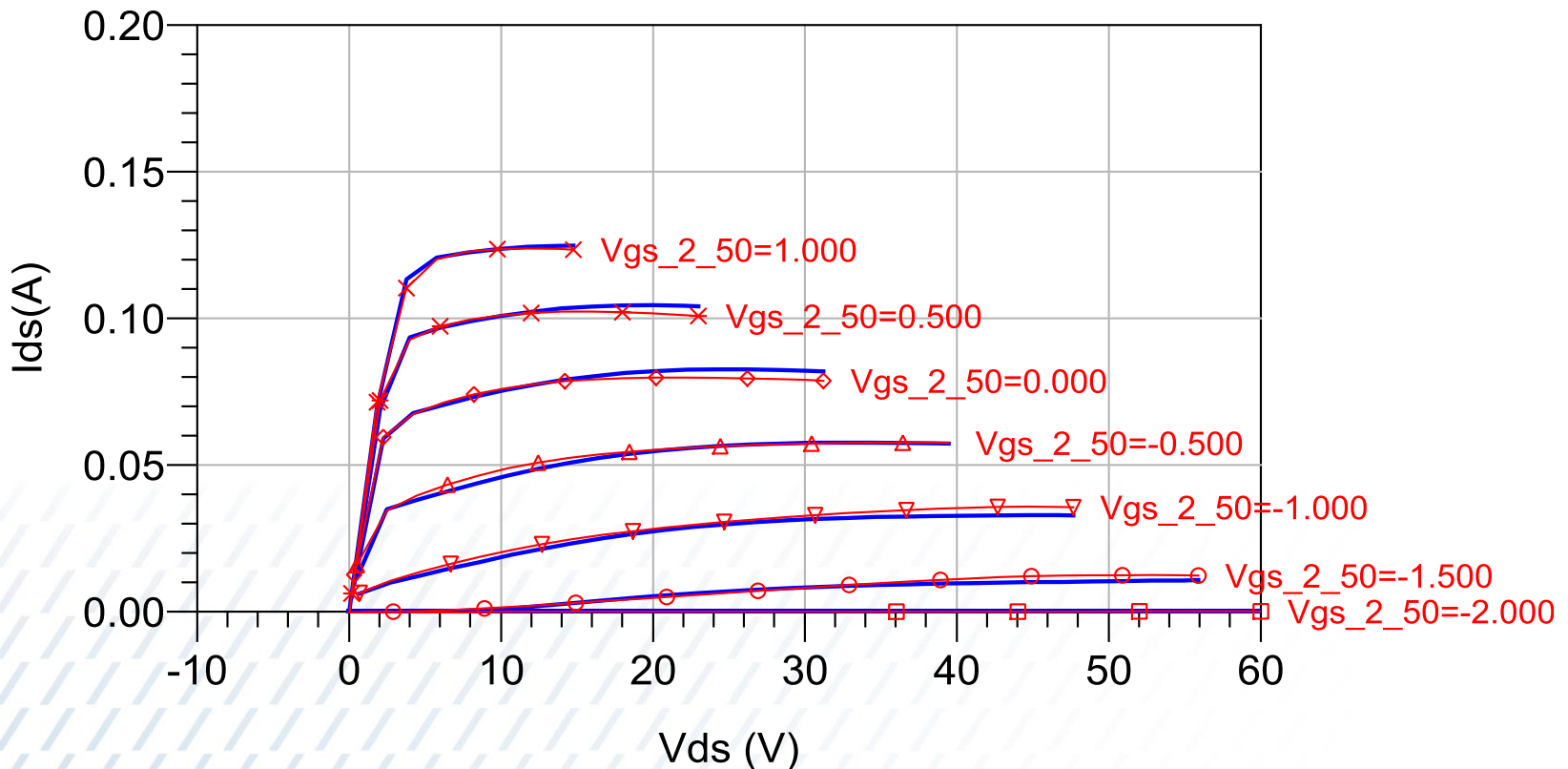
VAR
VAR46
index_step=2

Trap Validation

Transient Simulation

➤ Gate & Drain Trap measurement at 500 ns

- $V_{gs0} = -2 \text{ V}$ & $V_{ds0} = 50 \text{ V}$



> Pulsed IV

- Thermal Validation
- Id decreasing
- Trap Validation
- **Transient versus Self Heating Ratio**

> Pulsed S-Parameters

- 28 V – 20 mA/mm
- 50 V – 20 mA/mm

> Static S-Parameters

- Class C & Class AB at 28 V

> LoadPull Performances

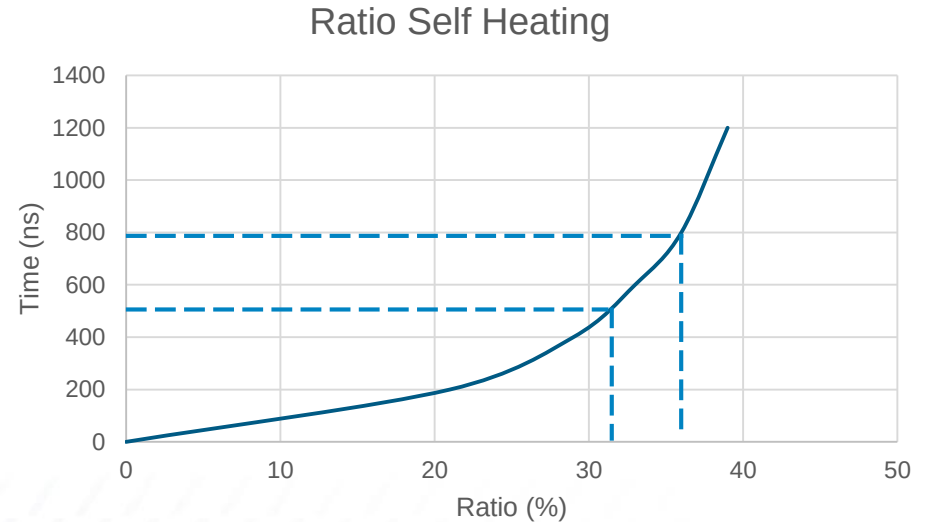
- 1.8 GHz at MXP & MXE - 28 V
- 3.8 GHz at MXP & MXE - 28 V

Transient versus Self Heating Ratio

S Parameters Conditions

> Ratio Self Heating

Selfheating %	Time (s)
0%	0.00E+00
21%	2.00E-07
29%	4.00E-07
33%	6.00E-07
36%	8.00E-07
39%	1.20E-06
63%	6.12E-05
68%	1.21E-04
71%	1.81E-04
75%	3.01E-04
77%	3.61E-04
79%	4.81E-04
82%	6.61E-04
94%	5.51E-03
100%	3.17E-02



S Par : 500 ns

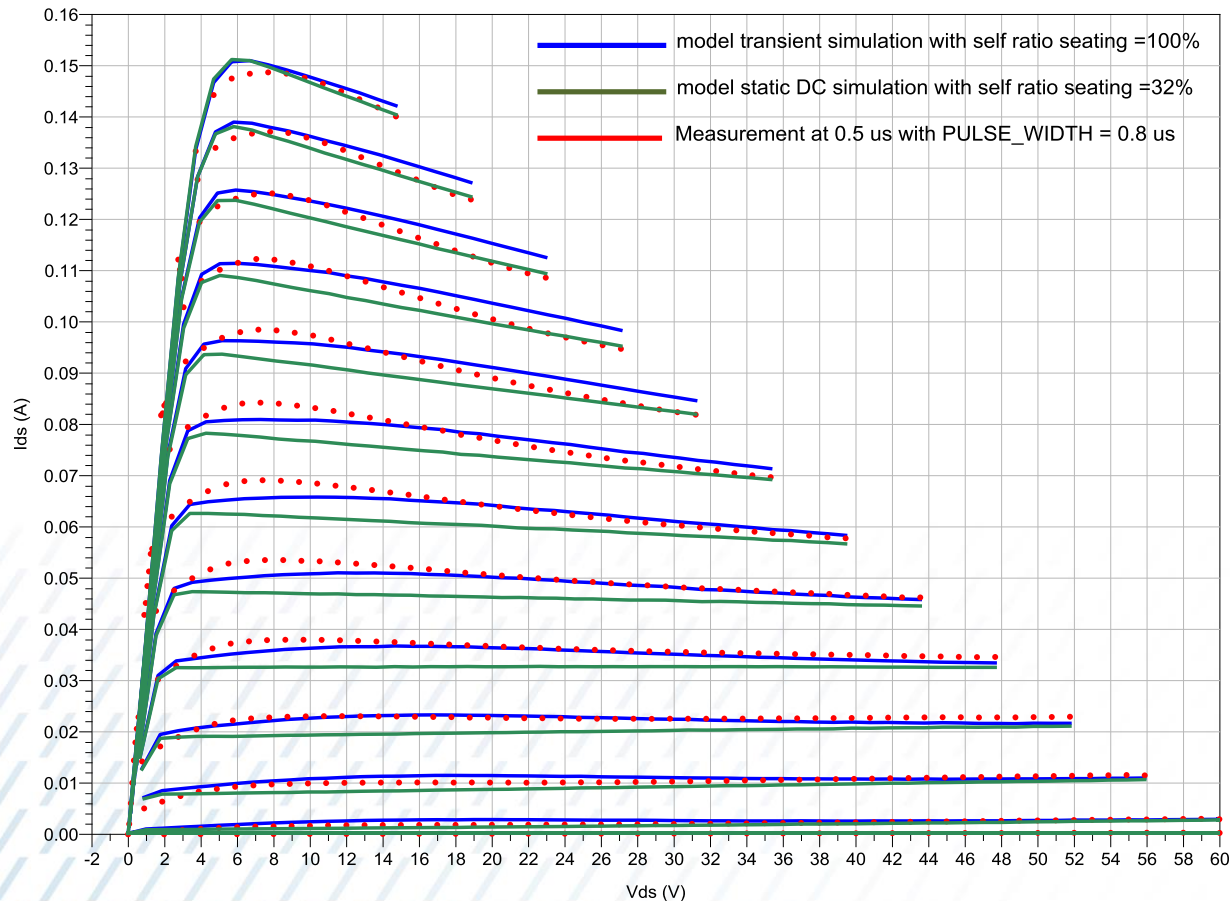
LoadPull : 800 ns

Transient versus Self Heating Ratio

S Parameters Conditions

➤ Self Heating Ratio = 32 % (from table)

- 25°C, Pulse Width = 0.8 us, Measurement at 0.5 us

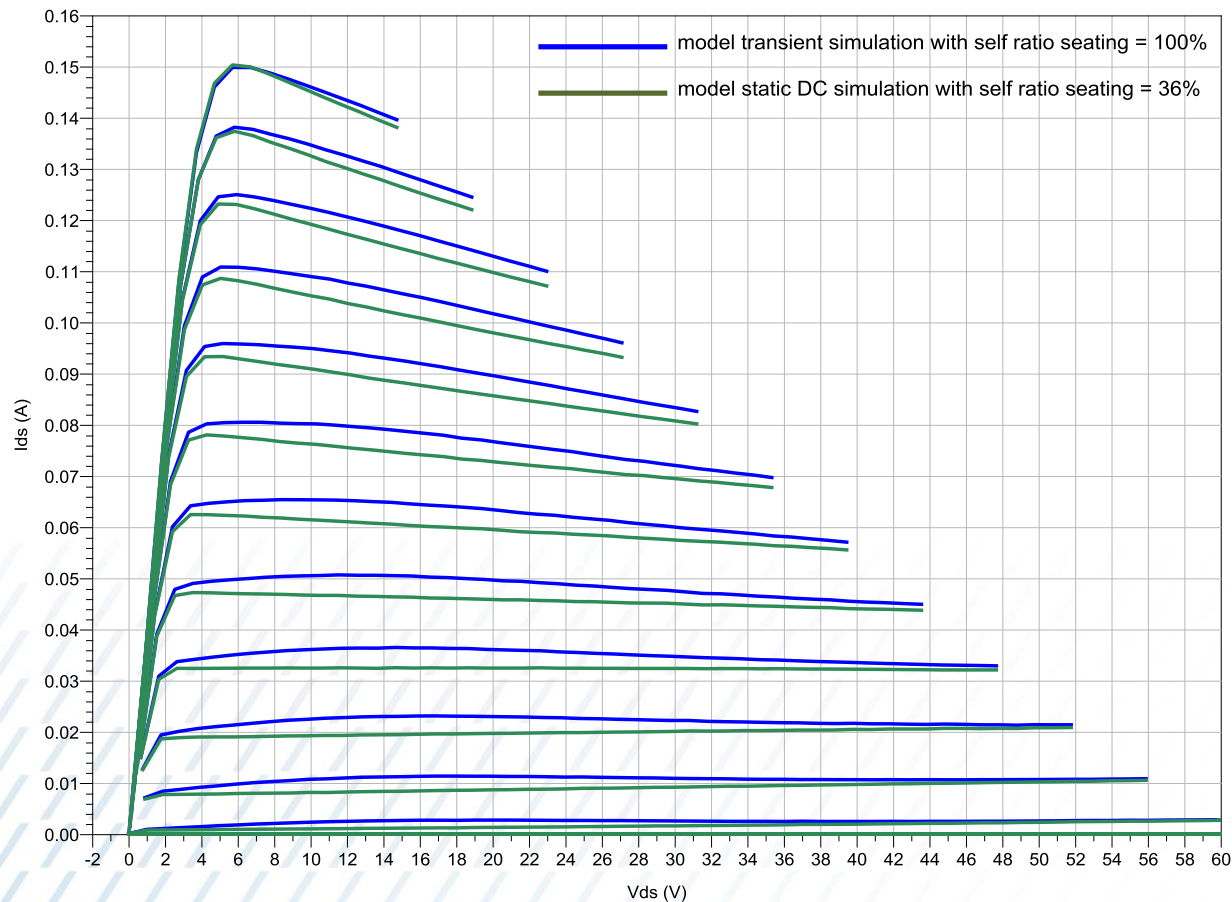


Transient versus Self Heating Ratio

LoadPull Conditions

➤ Self Heating Ratio = 36 % (from table)

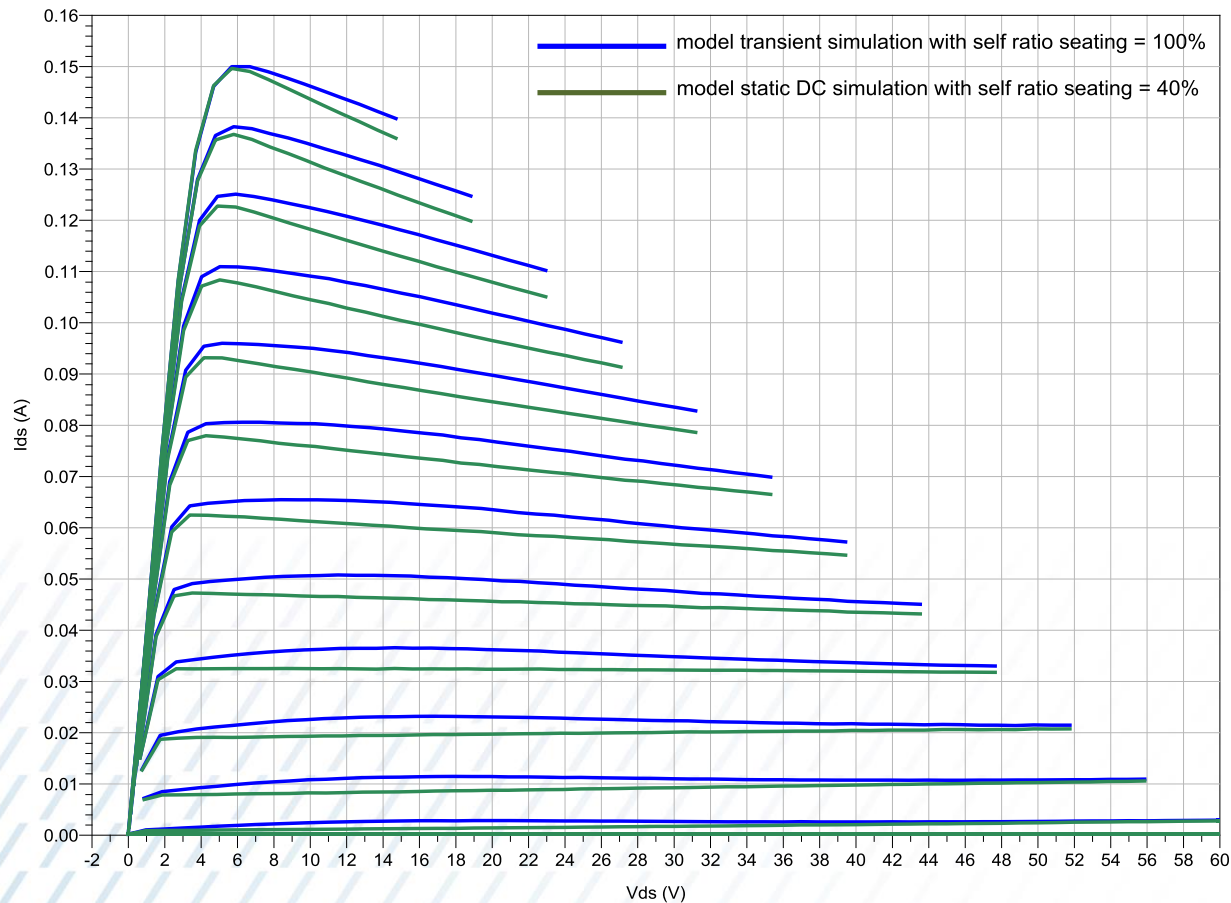
- 25°C, Pulse Width = 1 us, Measurement at 0.8 us



Transient versus Self Heating Ratio

LoadPull Conditions

- Self Heating Ratio = 40 % (as recommended)
 - 25°C, Pulse Width = 1 us, Measurement at 0.8 us



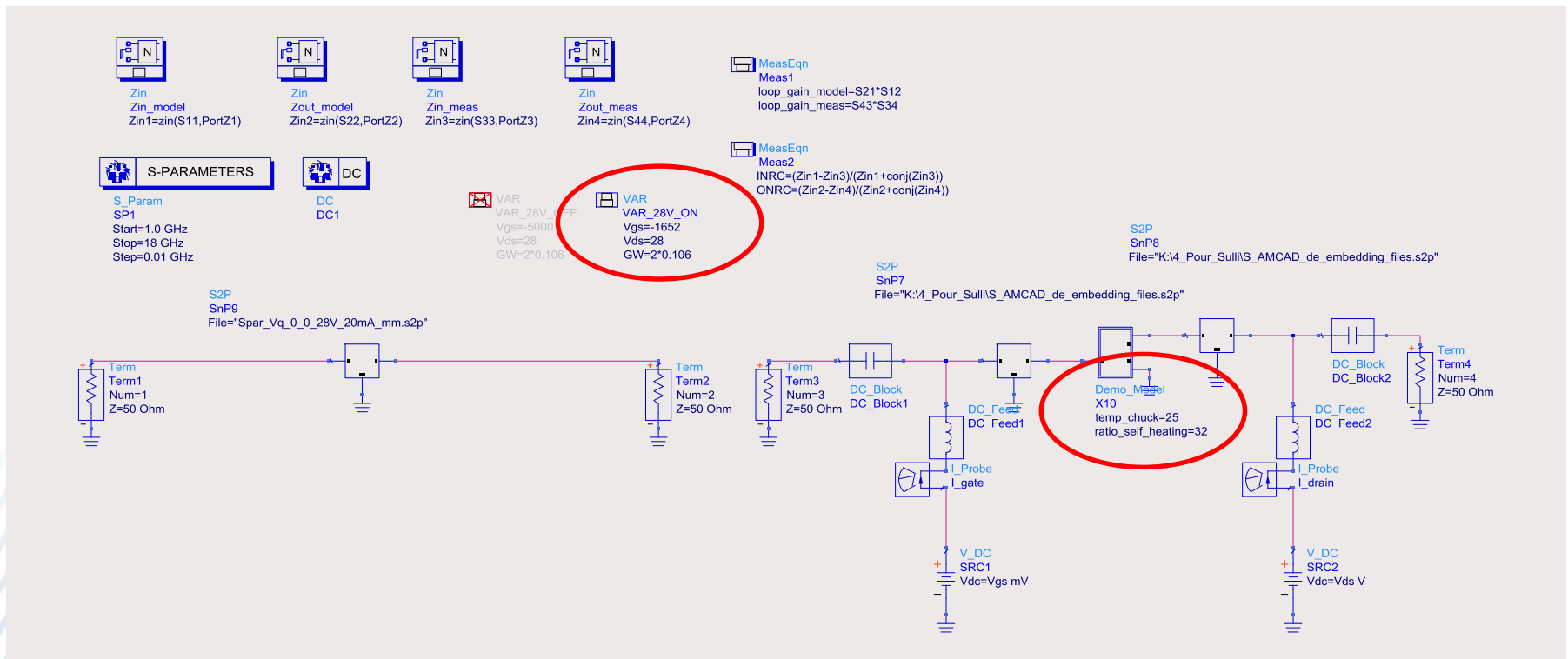
- Pulsed IV
 - Thermal Validation
 - Id decreasing
 - Trap Validation
 - Transient versus Self Heating Ratio
- **Pulsed S-Parameters**
 - **28 V – 20 mA/mm**
 - 50 V – 20 mA/mm
- Static S-Parameters
 - Class C & Class AB at 28 V
- LoadPull Performances
 - 1.8 GHz at MXP & MXE - 28 V
 - 3.8 GHz at MXP & MXE - 28 V

Pulsed S-Parameters

Class AB

➤ 1 measurement file

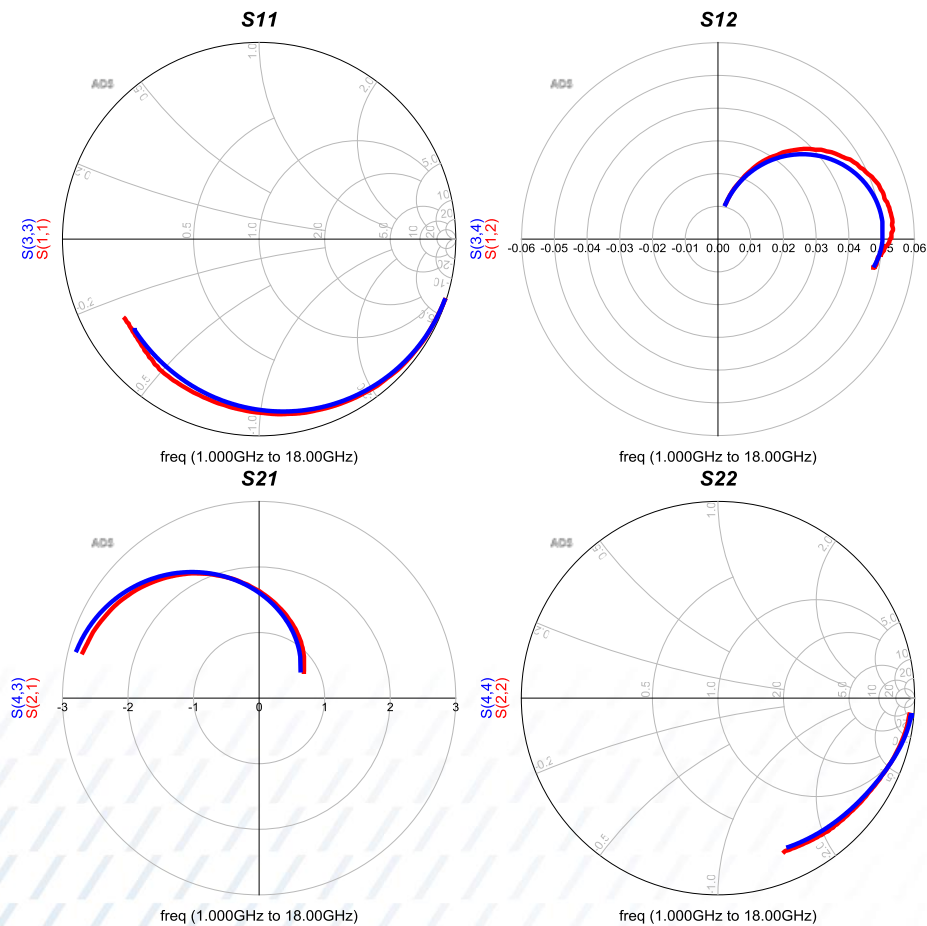
- "Spar_Vq_0_0_28V_20mA_mm_Deembedded.s2p"



Pulsed S-Parameters

Class AB

➤ S-Parameters

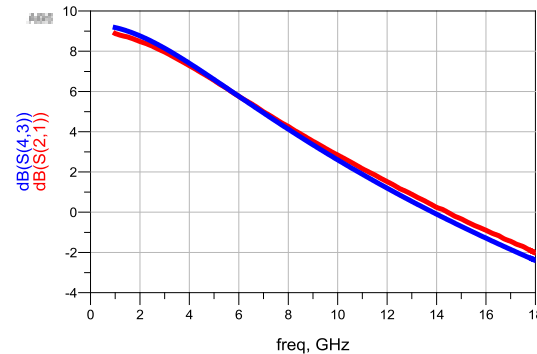


Pulsed S-Parameters

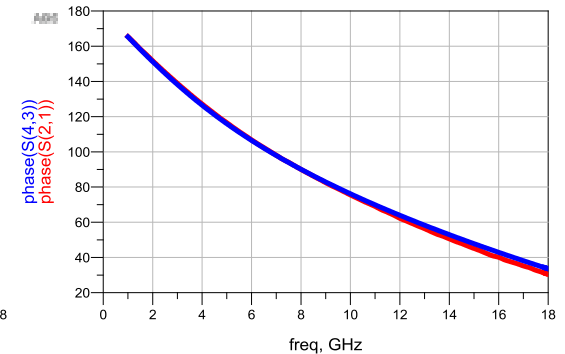
Class AB

➤ S-Parameters

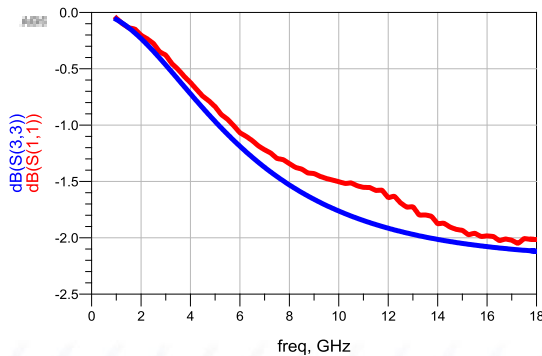
dB(S21)



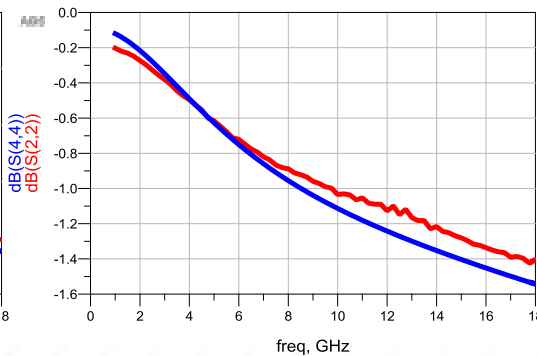
phase(S21)



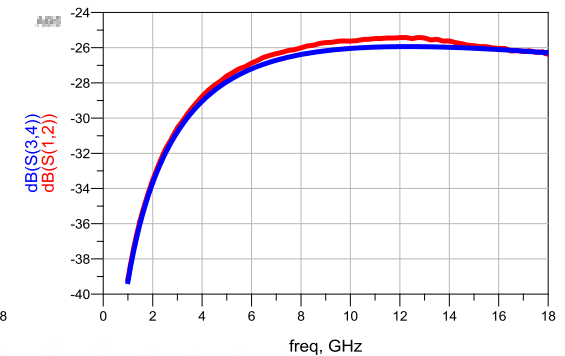
dB(S11)



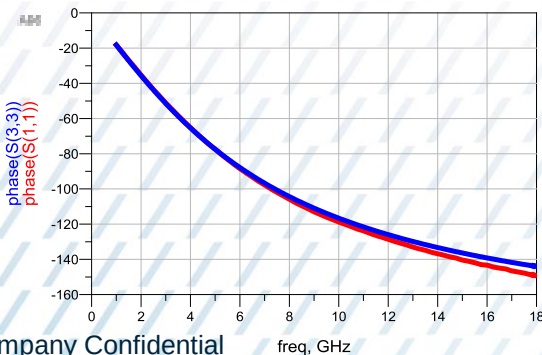
dB(S22)



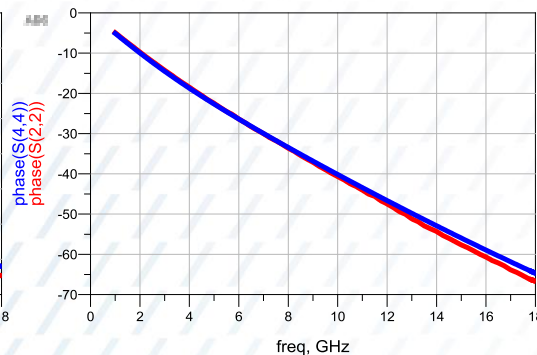
dB(S12)



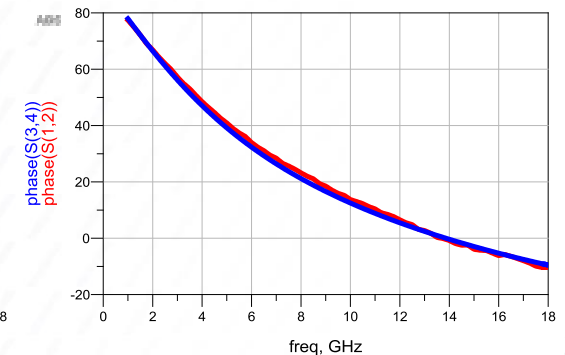
phase(S11)



phase(S22)



phase(S12)



Pulsed S-Parameters

Class AB

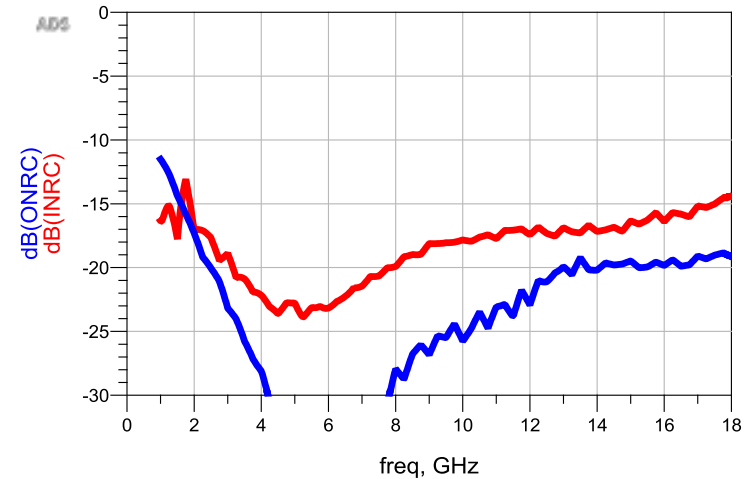
> S-Parameters

- $V_g = -1.645 \text{ V}$

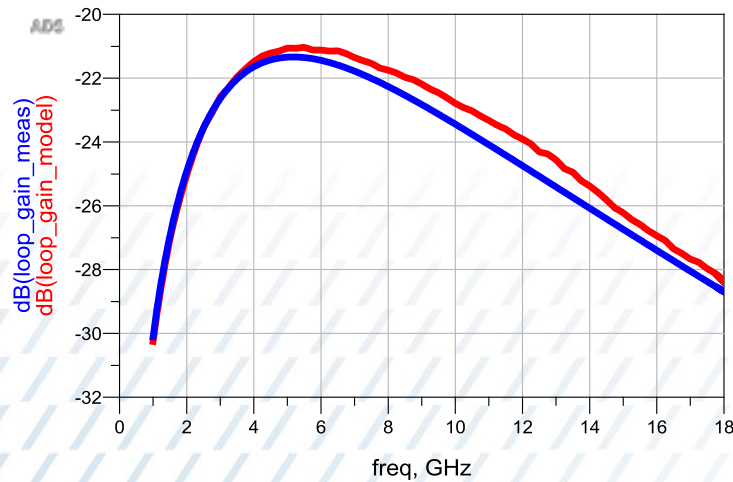
Eqn $I_{\text{Drain_mA_mm}} = 1000 * I_{\text{drain.i}} / (2 * 0.106)$

freq	I_drain.i	I_Drain_mA_mm
0.0000 Hz	4.250 mA	20.046

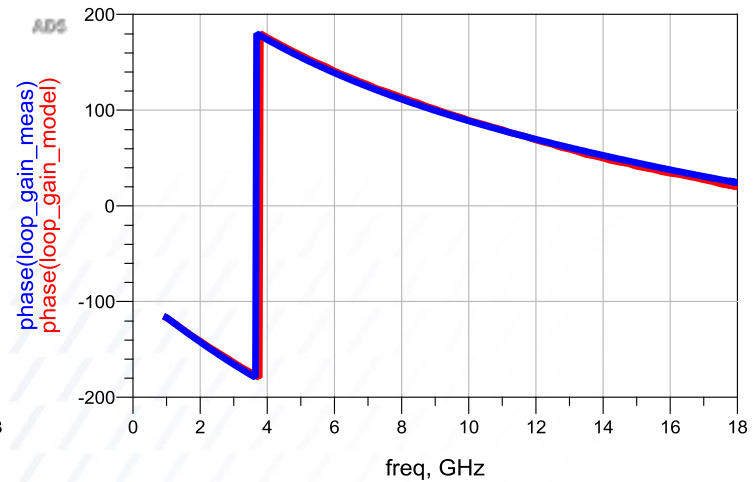
normalised reflection coefficients



dB(loop gain)



phase(loop gain)



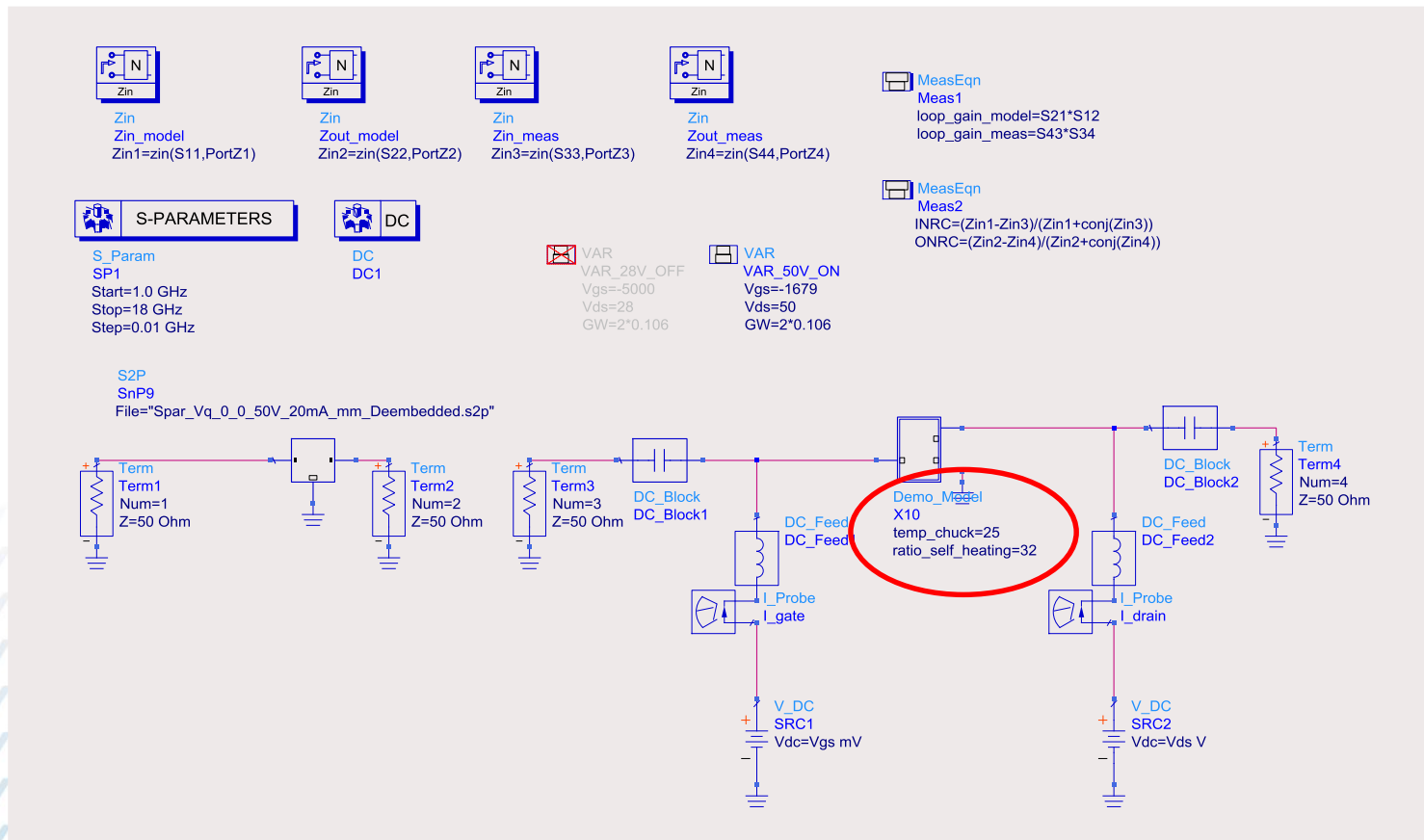
- Pulsed IV
 - Thermal Validation
 - Id decreasing
 - Trap Validation
 - Transient versus Self Heating Ratio
- **Pulsed S-Parameters**
 - 28 V – 20 mA/mm
 - **50 V – 20 mA/mm**
- Static S-Parameters
 - Class C & Class AB at 28 V
- LoadPull Performances
 - 1.8 GHz at MXP & MXE - 28 V
 - 3.8 GHz at MXP & MXE - 28 V

Pulsed S-Parameters

Class AB

➤ 1 measurement file

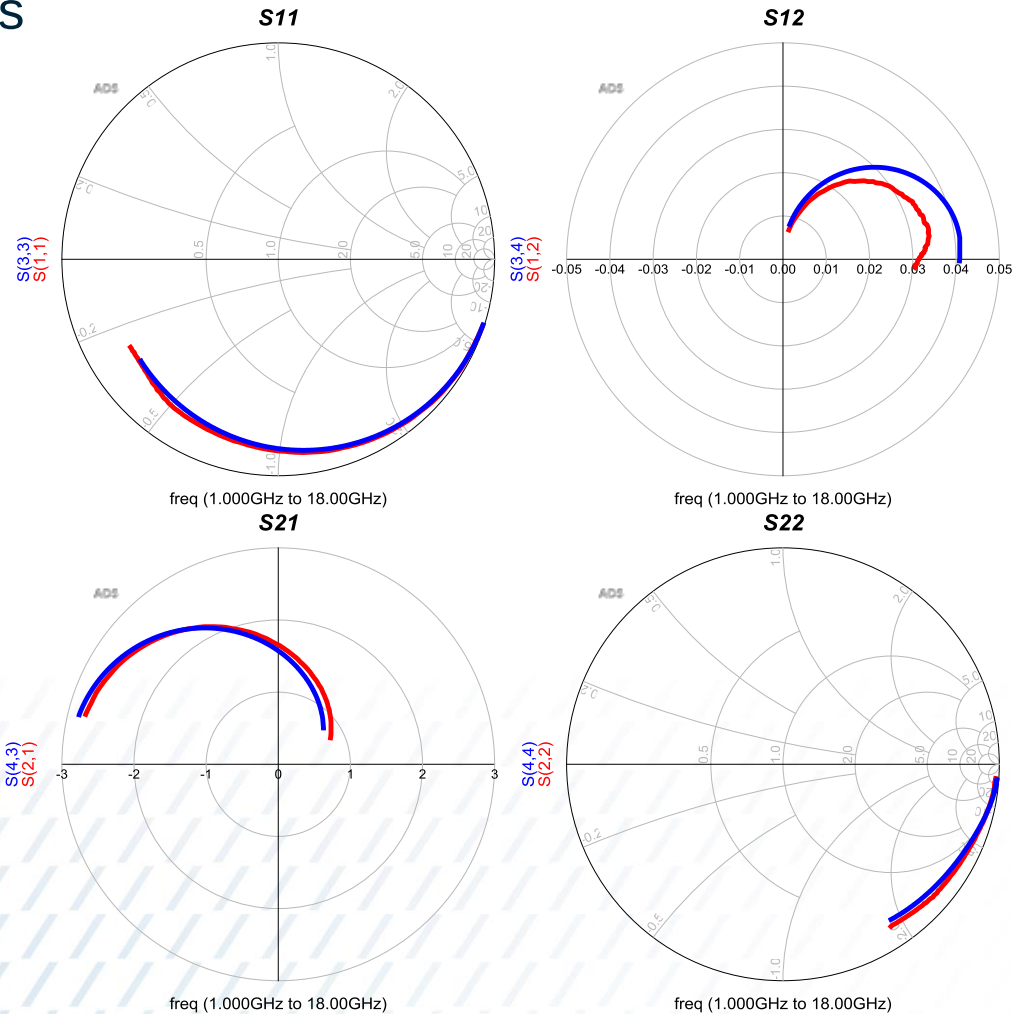
- “Spar_Vq_0_0_50V_20mA_mm_Deembedded.s2p”



Pulsed S-Parameters

Class AB

➤ S-Parameters

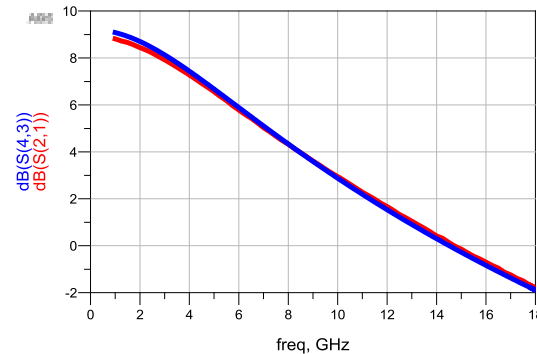


Pulsed S-Parameters

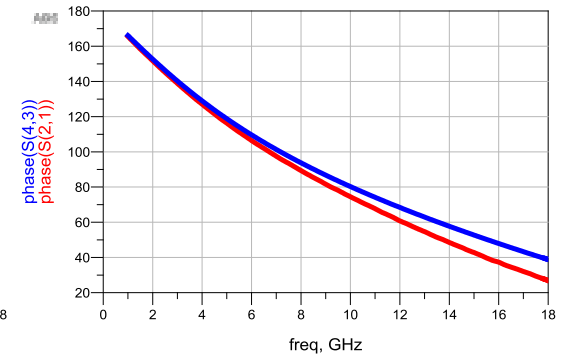
Class AB

➤ S-Parameters

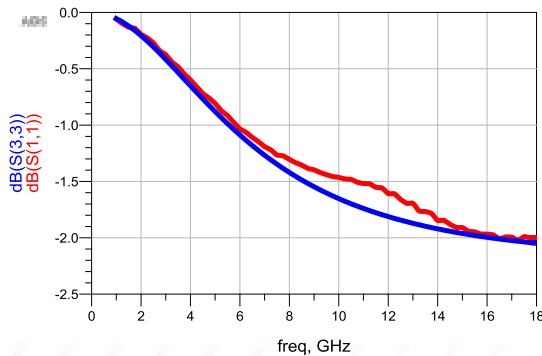
dB(S21)



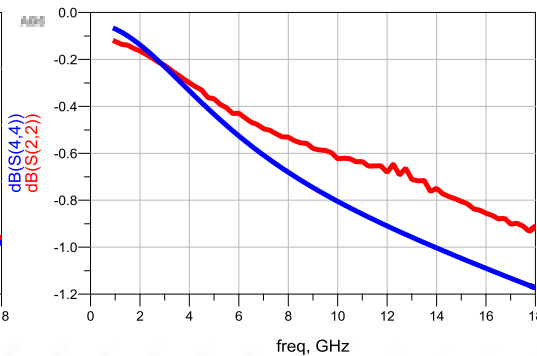
phase(S21)



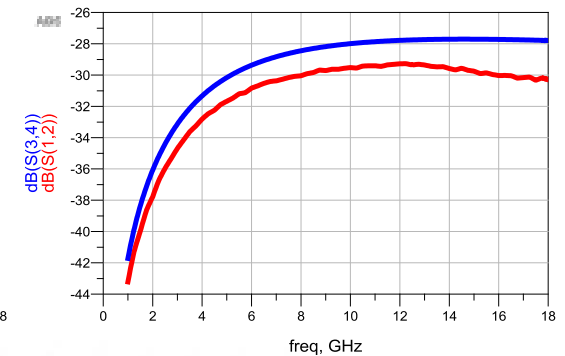
dB(S11)



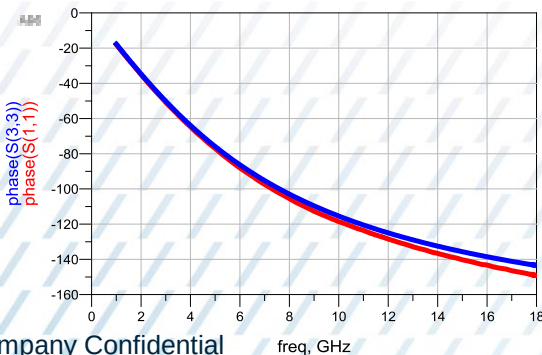
dB(S22)



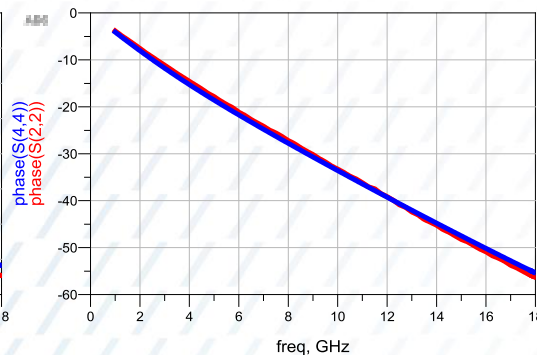
dB(S12)



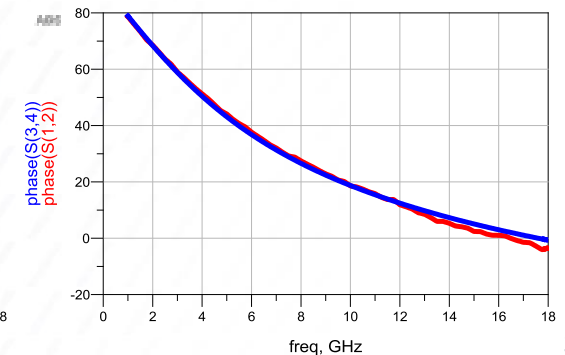
phase(S11)



phase(S22)



phase(S12)



Pulsed S-Parameters

Class AB

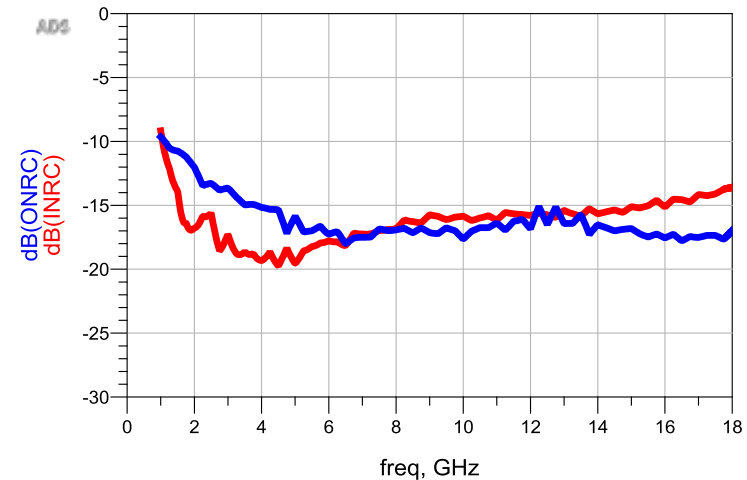
> S-Parameters

- $V_g = -1.679$ V

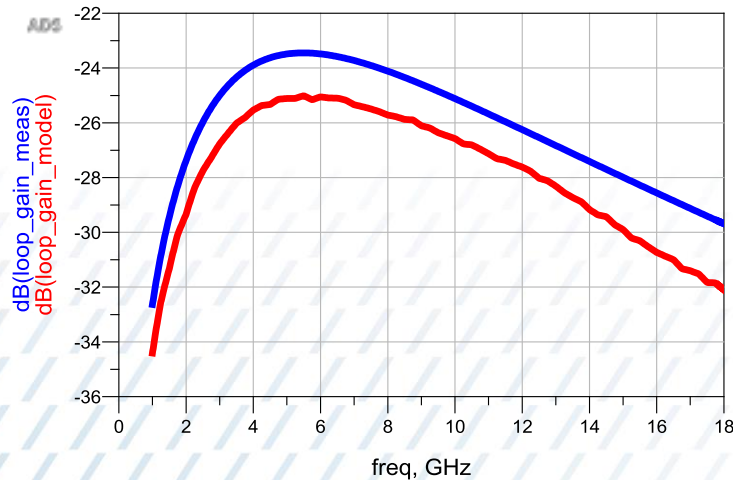
$$Eqn \ I_Drain_mA_mm = 1000 * I_drain.i / (2 * 0.106)$$

freq	I_drain.i	I_Drain_mA_mm
0.0000 Hz	4.251 mA	20.054

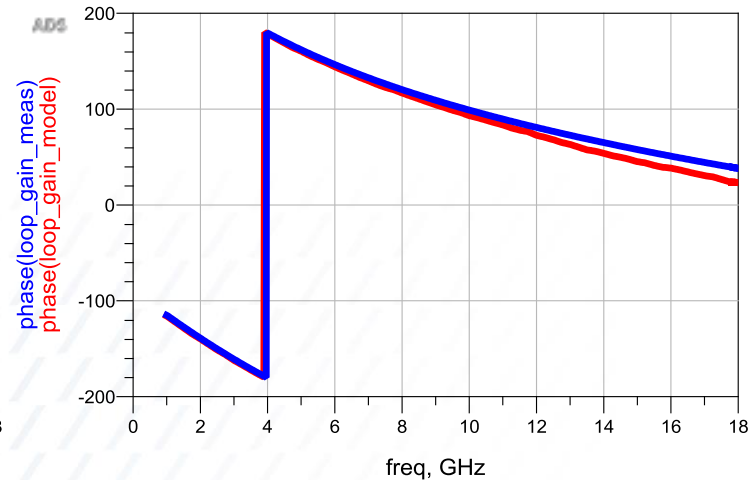
normalised reflection coefficients



dB(loop gain)



phase(loop gain)



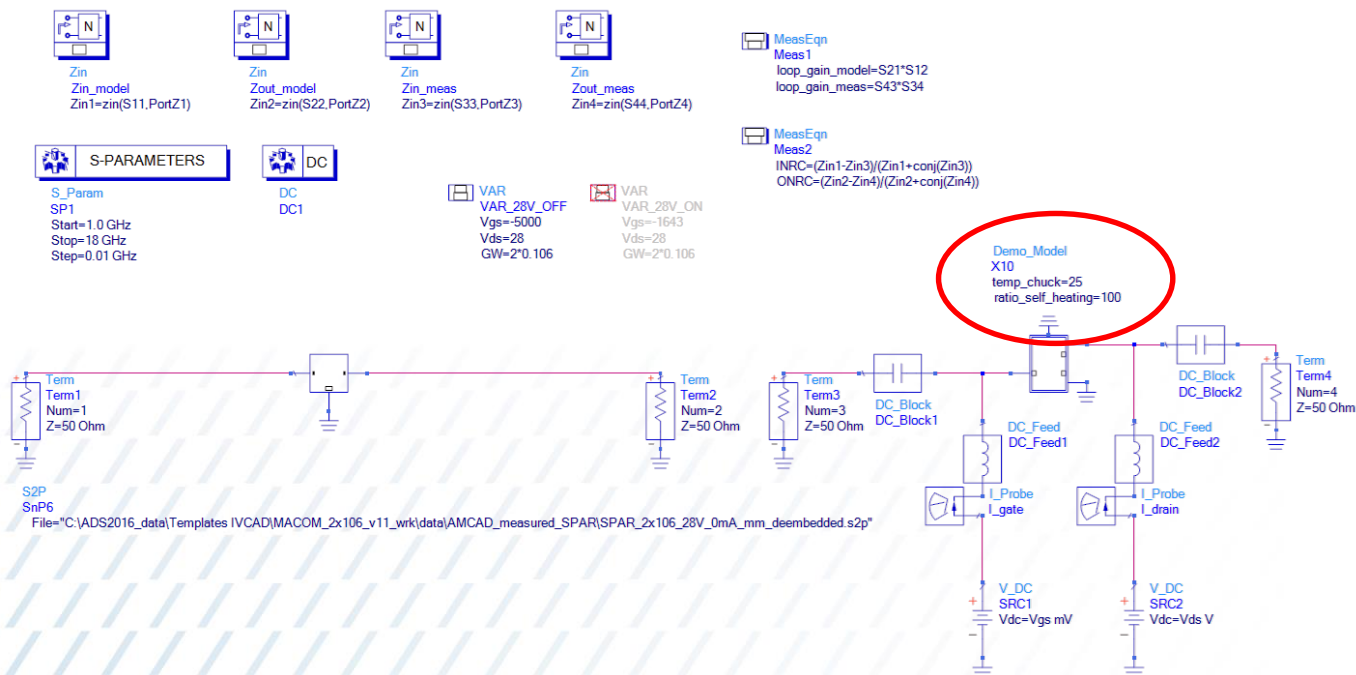
- Pulsed IV
 - Thermal Validation
 - Id decreasing
 - Trap Validation
 - Transient versus Self Heating Ratio
- Pulsed S-Parameters
 - 28 V – 20 mA/mm
 - 50 V – 20 mA/mm
- **Static S-Parameters**
 - **Class C & Class AB at 28 V**
- LoadPull Performances
 - 1.8 GHz at MXP & MXE - 28 V
 - 3.8 GHz at MXP & MXE - 28 V

Static S-Parameters

Model versus Measurement

➤ 2 measurements files

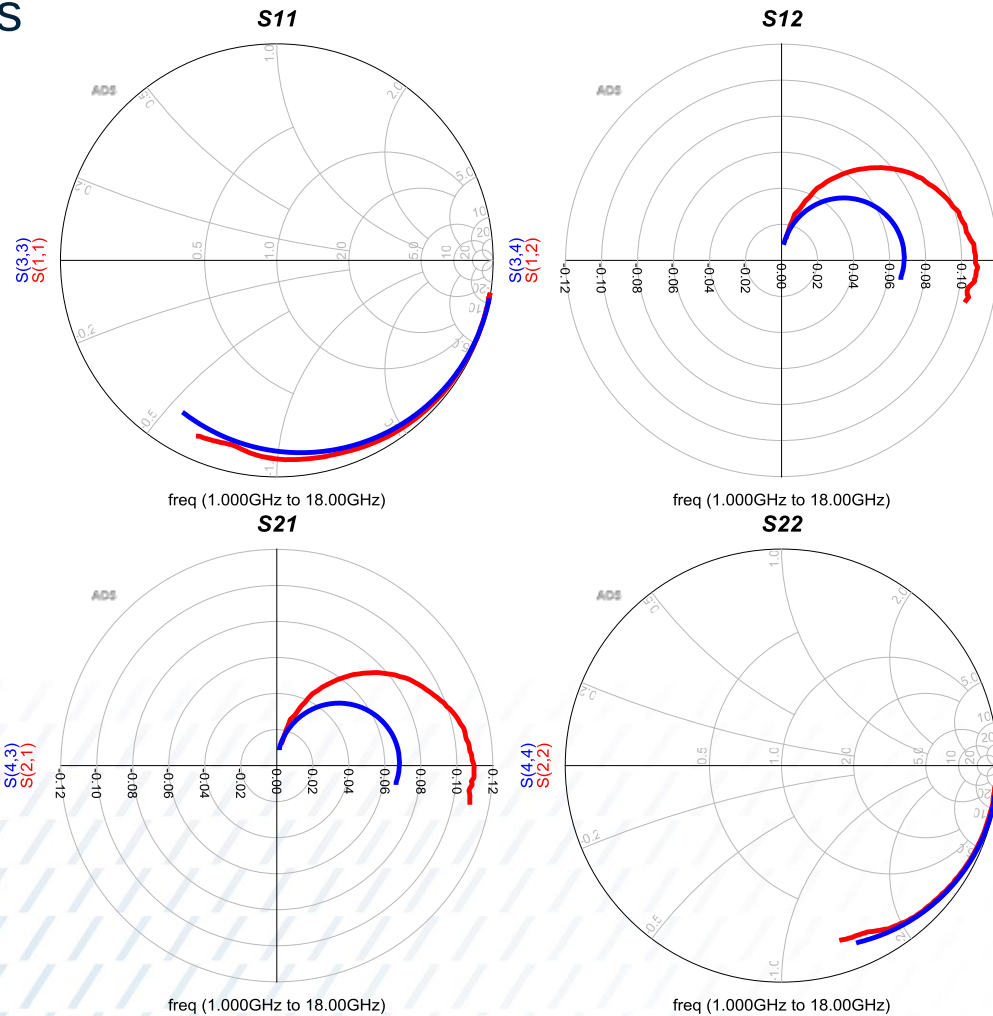
- SPAR_2x106_28V_0mA_mm_deembedded.s2p
- SPAR_2x106_28V_20mA_mm_deembedded.s2p



Static S-Parameters

SPAR_2x106_28V_0mA_mm_deembedded.s2p

➤ S-Parameters

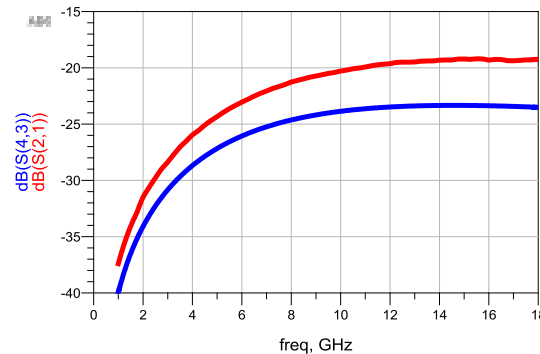


Static S-Parameters

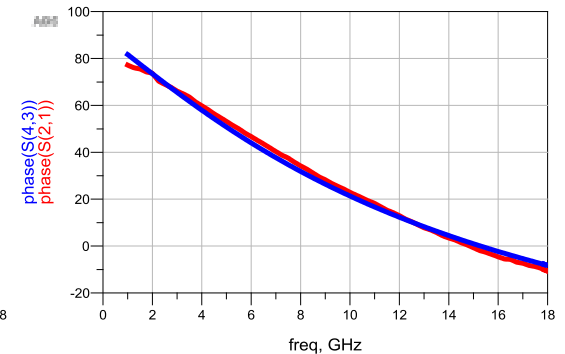
SPAR_2x106_28V_0mA_mm_deembedded.s2p

➤ S-Parameters

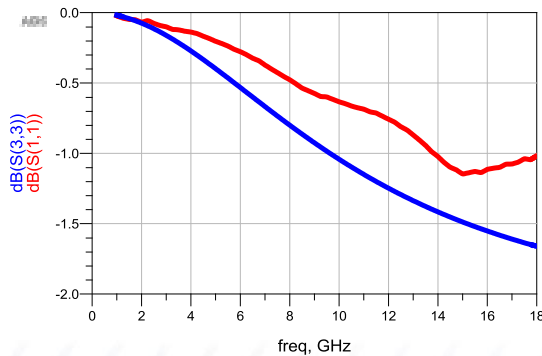
dB(S21)



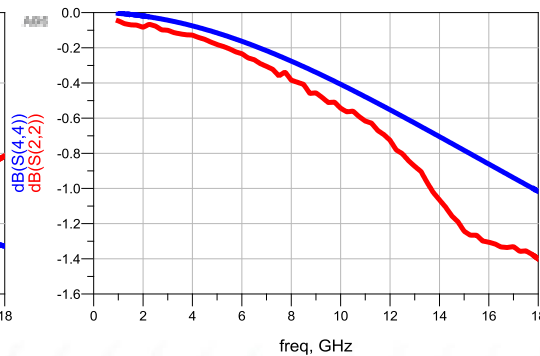
phase(S21)



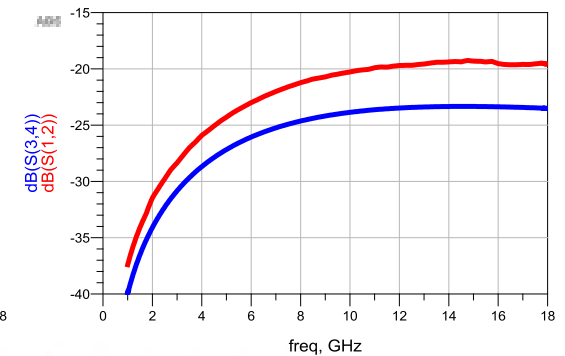
dB(S11)



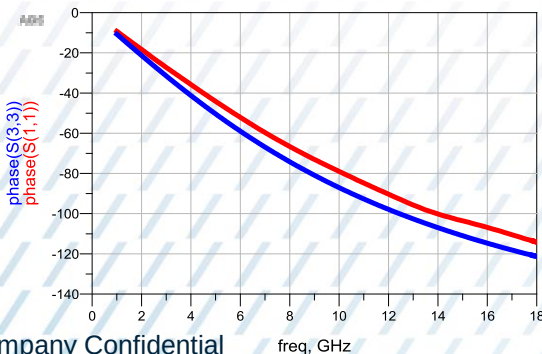
dB(S22)



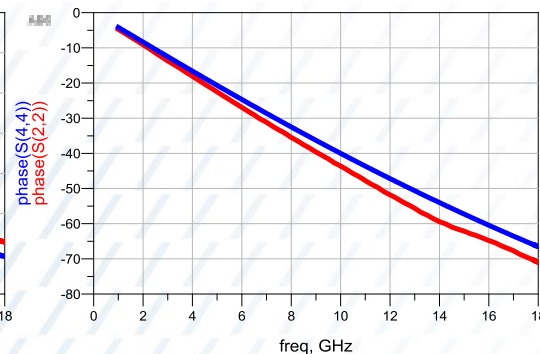
dB(S12)



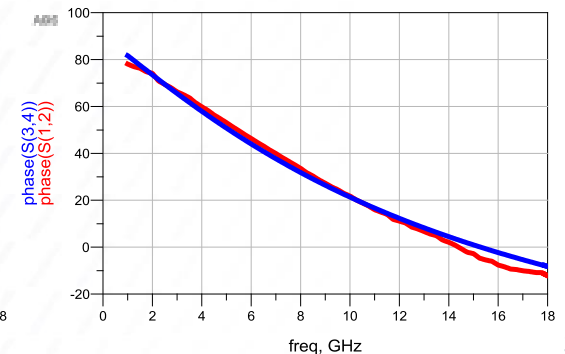
phase(S11)



phase(S22)



phase(S12)



Static S-Parameters

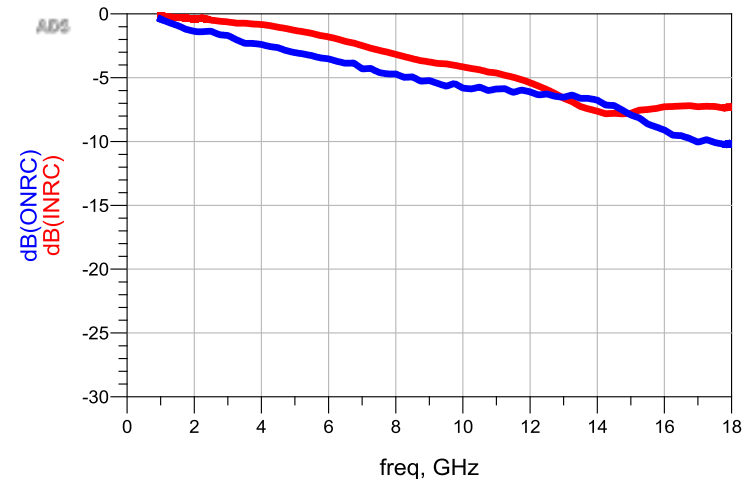
SPAR_2x106_28V_0mA_mm_deembedded.s2p

> S-Parameters

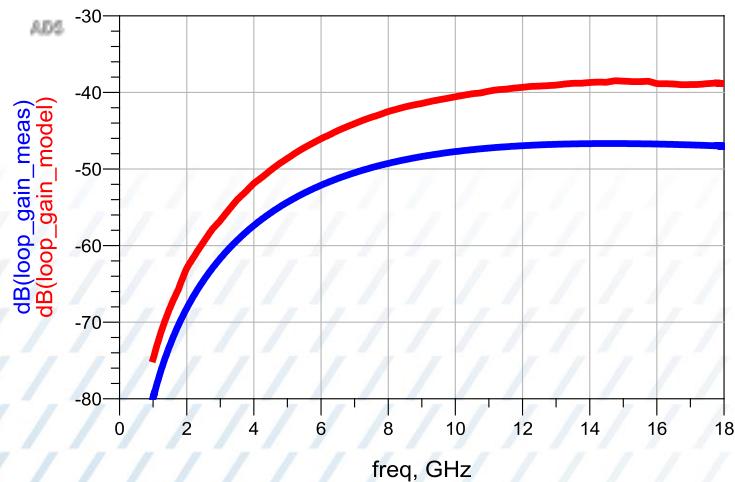
Eqn $I_{\text{Drain_mA_mm}} = 1000 * I_{\text{drain.i}} / (2 * 0.106)$

freq	I_drain.i	I_Drain_mA_mm
0.0000 Hz	188.1 uA	0.887

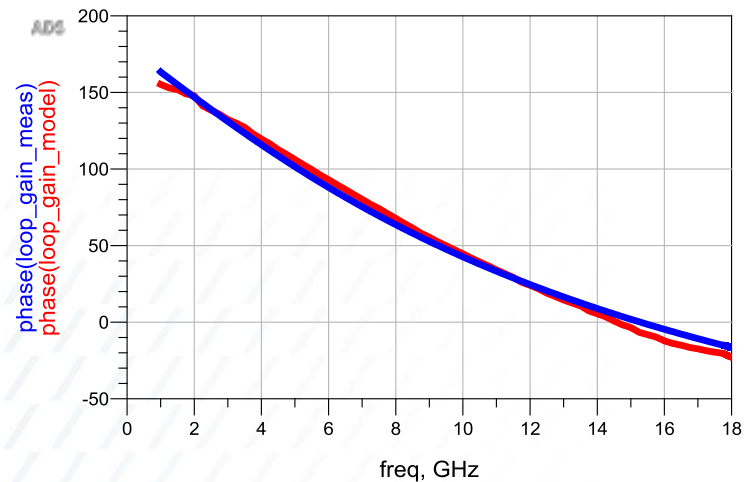
normalised reflection coefficients



dB(loop gain)



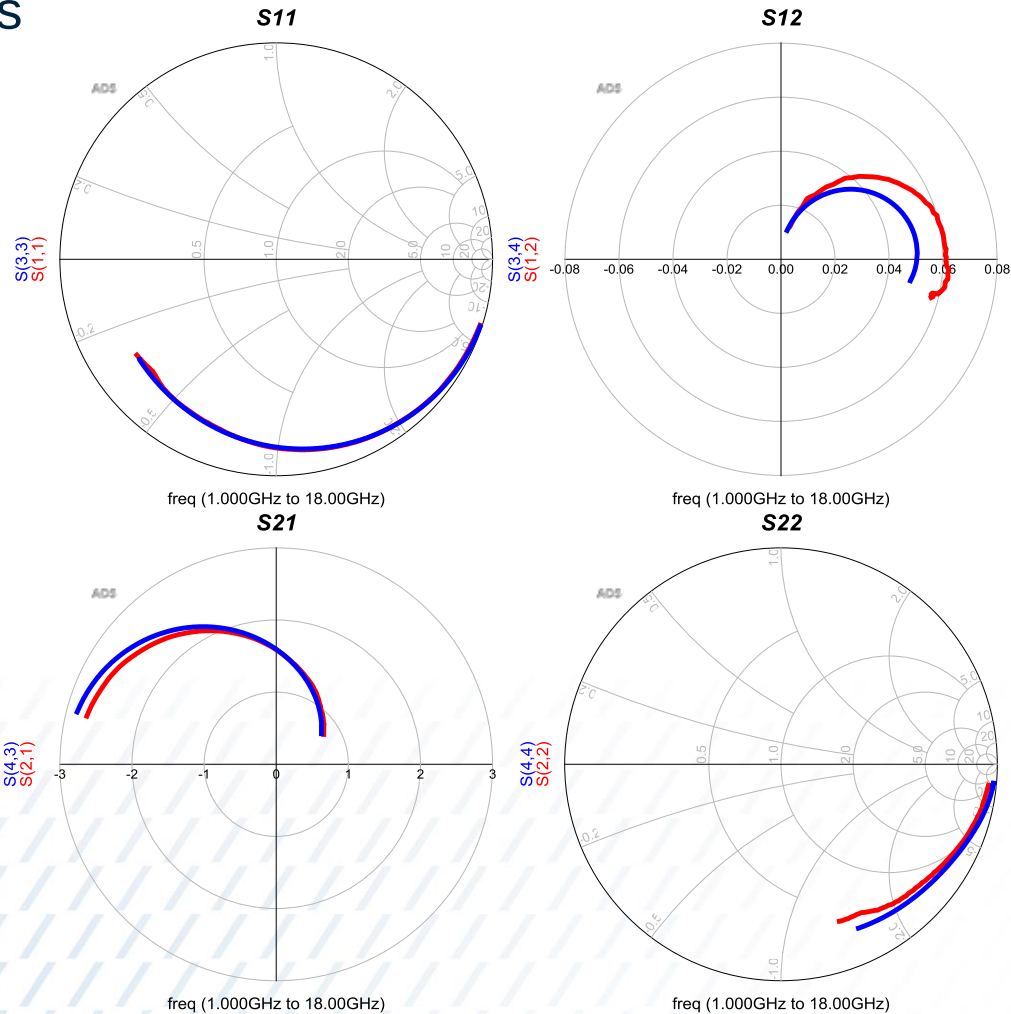
phase(loop gain)



Static S-Parameters

SPAR_2x106_28V_20mA_mm_deembedded.s2p

➤ S-Parameters

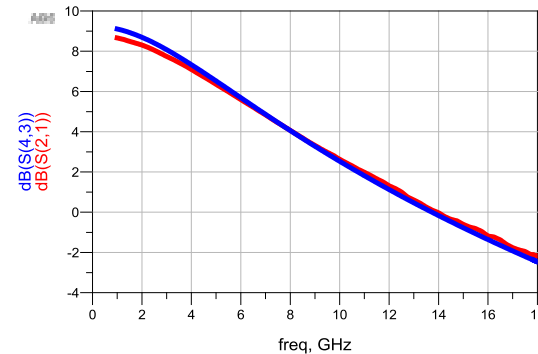


Static S-Parameters

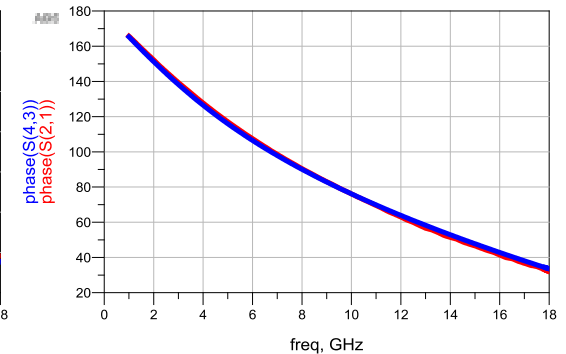
SPAR_2x106_28V_20mA_mm_deembedded.s2p

➤ S-Parameters

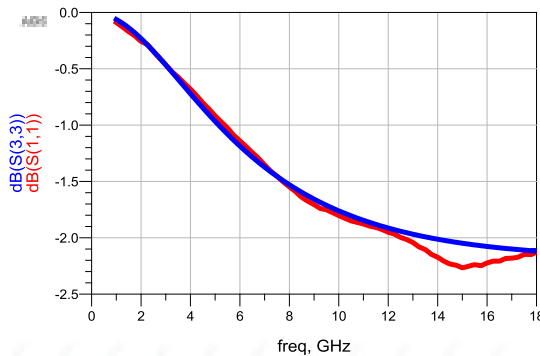
dB(S21)



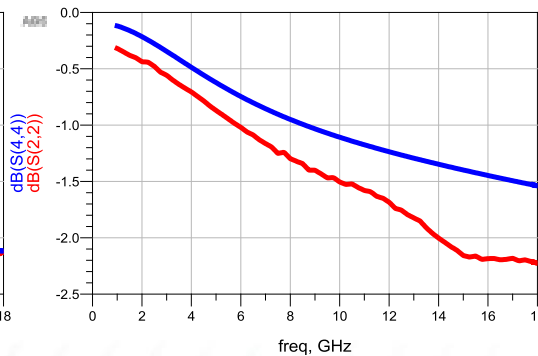
phase(S21)



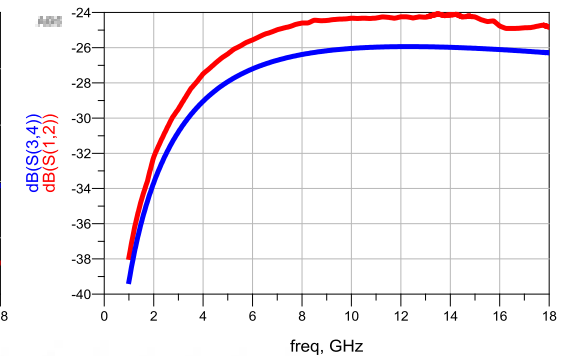
dB(S11)



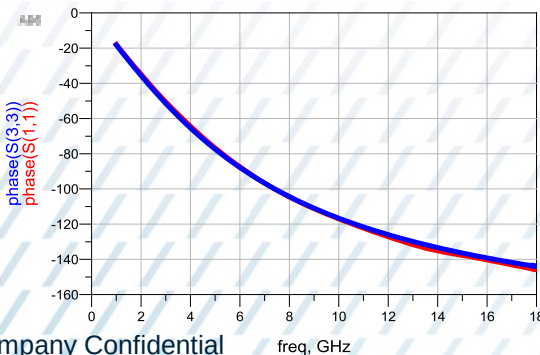
dB(S22)



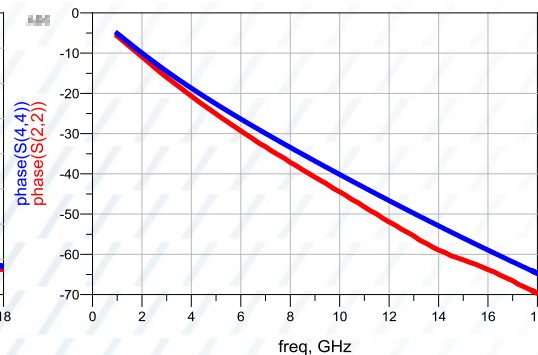
dB(S12)



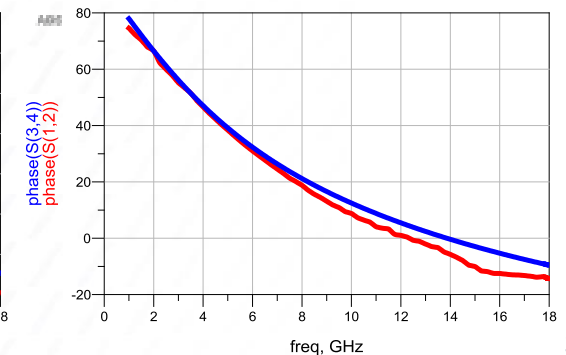
phase(S11)



phase(S22)



phase(S12)



Static S-Parameters

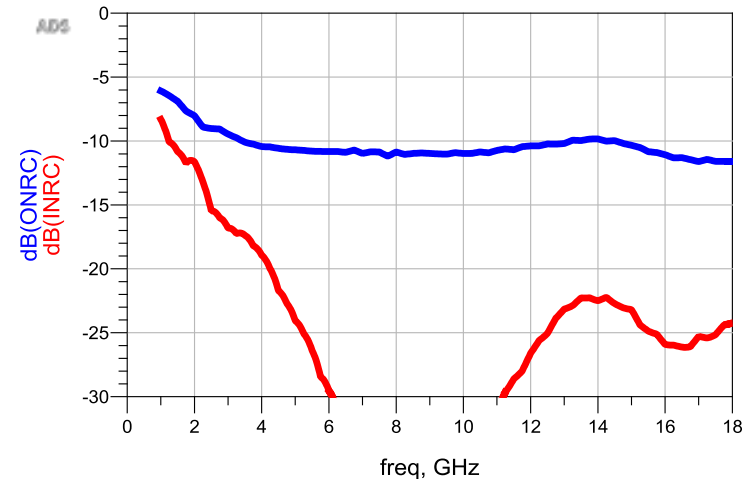
SPAR_2x106_28V_20mA_mm_deembedded.s2p

> S-Parameters

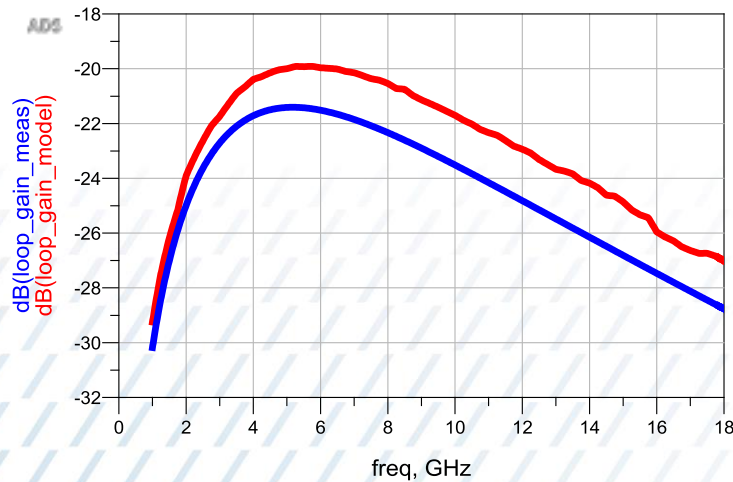
Eqn I_Drain_mA_mm=1000*I_drain.i/(2*0.106)

freq	I_drain.i	I_Drain_mA_mm
0.0000 Hz	4.249 mA	20.044

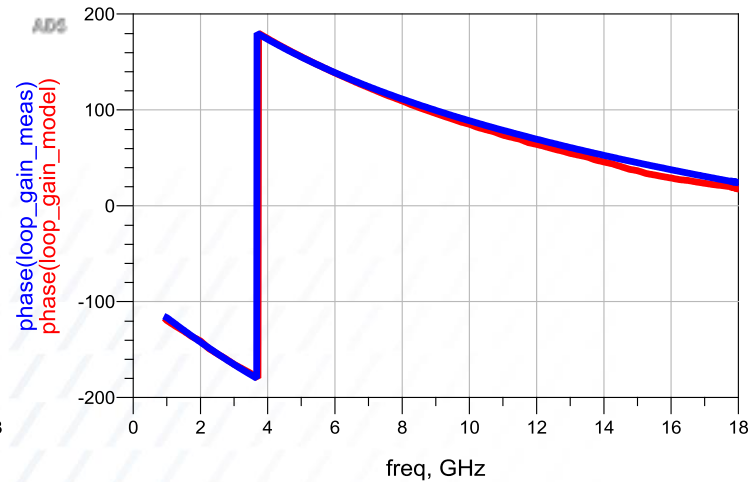
normalised reflection coefficients



dB(loop gain)



phase(loop gain)



- Pulsed IV
 - Thermal Validation
 - Id decreasing
 - Trap Validation
 - Transient versus Self Heating Ratio
- Pulsed S-Parameters
 - 28 V – 20 mA/mm
 - 50 V – 20 mA/mm
- Static S-Parameters
 - Class C & Class AB at 28 V
- **LoadPull Performances**
 - **1.8 GHz at MXP & MXE - 28 V**
 - 3.8 GHz at MXP & MXE - 28 V

> Measurement Files

- 1.8 GHz
 - 2bis_Step1_PS_IS_Pout_Sample_D2_2x106 μ m_Vq_28V_20mA_mm_f_1-8GHz_loadf0swept_2f03f0loadsource_50_Deembedded.cst
 - 3bis2_Step1_PS_IS_PAE_Sample_D2_2x106 μ m_Vq_28V_20mA_mm_f_1-8GHz_loadf0swept_2f03f0loadsource_50_Demebedded.cst
- 3.8 GHz
 - 2_Step1_PS_IS_Pout_Sample_D2_2x106 μ m_Vq_28V_20mA_mm_f_3-8GHz_loadf0swept_2f03f0loadsource_50_Deembedded.cst
 - 3bis_Step1_PS_IS_PAE_Sample_D2_2x106 μ m_Vq_28V_20mA_mm_f_3-8GHz_loadf0swept_2f03f0loadsource_50_Deembedded.cst

LoadPull Performances

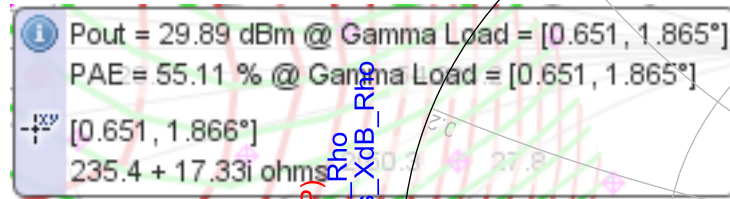
1.8 GHz at MXP – 28 V

— Measurement

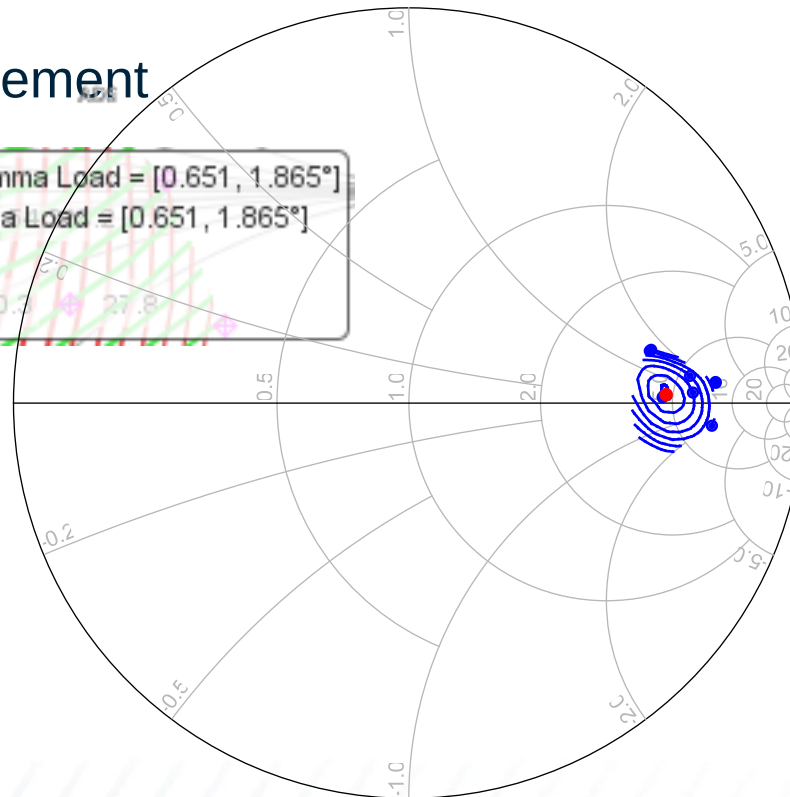
— Simulation

MACOM™

➤ From measurement



Same for AMCAD



30.200
30.000
29.800
29.600

Zload_at mPdel_vs_PAE	228.889 + j11.111
PAE_at mPdel_vs_PAE	58.960
...dBm_at mPdel_vs_PAE	30.771
Gain_at mPdel_vs_PAE	20.548

$$\text{Eqn } Z_{\text{MXP}} = 235.4 + j*17.33$$

$$\text{Eqn } Z_{\text{MXP_Simulation}} = \text{stoz}(\text{mPdel_vs_PAE_Rho})$$

$$\text{Eqn } \text{Coeff_Norm} = \text{db}((Z_{\text{MXP}} - Z_{\text{MXP_Simulation}}) / (Z_{\text{MXP}} + \text{conj}(Z_{\text{MXP_Simulation}})))$$

Coeff_Norm
-34.248

LoadPull Performances

1.8 GHz at MXP – 28 V

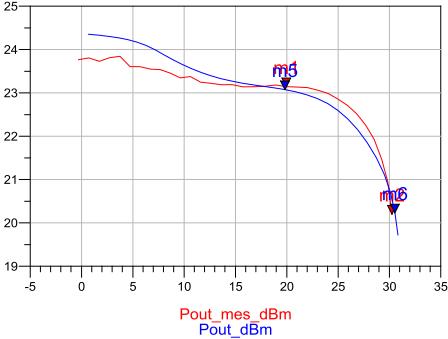
— Measurement
— Simulation



➤ Power Sweep Comparison

Cf Summary Table

GP vs OUTPUT POWER



m1
Pout_mes_dBm=19.955
plot_vs(PowerGain_mes_dB, Pout_mes_dBm)=23.142
idx_bloc=4.000

m5
Pout_dBm=19.807
plot_vs(T_LOAD_PULL_1p8_GHz_MXP_v1..Gp, T_LOAD_PULL_1p8_GHz_MXP_v1..Pout_dBm)=23.075
idx_bloc=1.000

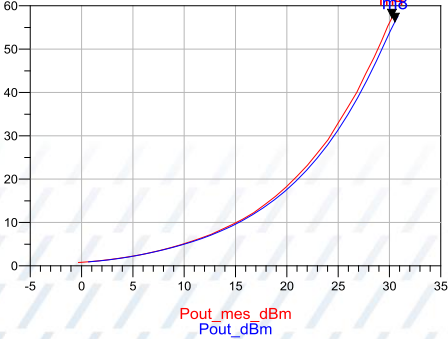
m2
Pout_mes_dBm=30.243
plot_vs(PowerGain_mes_dB, Pout_mes_dBm)=20.204
idx_bloc=4.000

m6
Pout_dBm=30.527
plot_vs(T_LOAD_PULL_1p8_GHz_MXP_v1..Gp, T_LOAD_PULL_1p8_GHz_MXP_v1..Pout_dBm)=20.233
idx_bloc=1.000

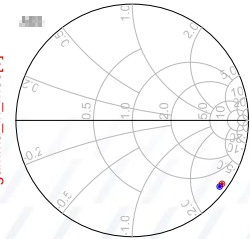
m4
Pout_mes_dBm=30.243
plot_vs(EFF_mes, Pout_mes_dBm)=57.196
idx_bloc=4.000

m8
Pout_dBm=30.527
plot_vs(real(T_LOAD_PULL_1p8_GHz_MXP_v1..Eff), T_LOAD_PULL_1p8_GHz_MXP_v1..Pout_dBm)=56.29
idx_bloc=1.000

EFF vs INPUT POWER

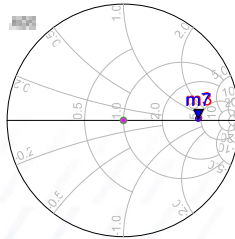


T_LOAD_PULL_1p8_GHz_MXP_v1..S(3,3)[0]
gamma_in_mes[0]



idx_bloc (4.000 to 4.000)

T_LOAD_PULL_1p8_GHz_MXP_v1..S(2,2)[0]
GLoad[0]
GLoad[20]
GLoad[0]



index_point (1.000 to 36.000)

m3
index_point=36.000
GLoad[0]=0.651 / 1.865
idx_bloc=4.000
impedance = 235.418 + j17.330

m7
indep(m7)=0
T_LOAD_PULL_1p8_GHz_MXP_v1..S(2,2)[0]=0.643 / 1.329
impedance = 229.407 + j11.663

idx_bloc	INRC
4.000	-11.454

idx_bloc	ONRC
4.000	-34.703

LoadPull Performances

1.8 GHz at MXE – 28 V

— Measurement
— Simulation

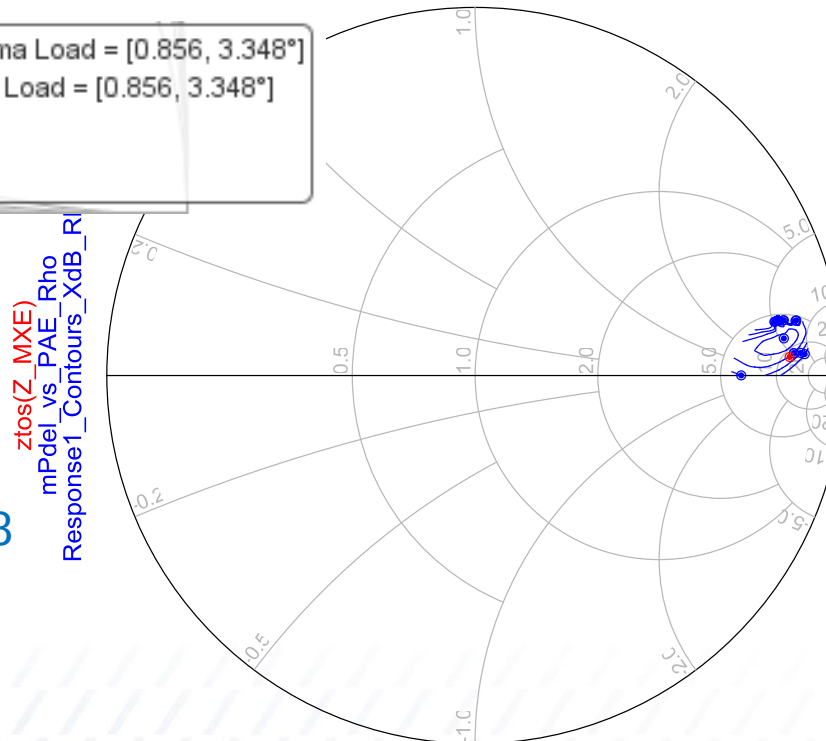
MACOM™

> From measurement

Pout = 27.64 dBm @ Gamma Load = [0.856, 3.348°]
 PAE = 61.45 % @ Gamma Load = [0.856, 3.348°]
 [0.856, 3.349°]
 563.5 + 210.2i ohms

AMCAD : 478+j*158

I do not find this impedance in the measurement file ..



Maximum & Contour levels, %:

72.659
72.000
70.000
68.000
66.000
64.000
62.000
60.000

Zload_at_mPdel_vs_PAE
395.556 + j277.778
PAE_at_mPdel_vs_PAE
67.492
...el_dBm_at_mPdel_vs_PAE
28.189
Gain_at_mPdel_vs_PAE
21.914

Eqn **Z_MXE=563+j*210**

Eqn Z_MXE_Simulation=stoz(mPdel_vs_PAE_Rho)

Eqn Coeff_Norm=db((Z_MXE-Z_MXE_Simulation)/(Z_MXE+conj(Z_MXE_Simulation)))

Coeff_Norm
-14.518

LoadPull Performances

1.8 GHz at MXE – 28 V

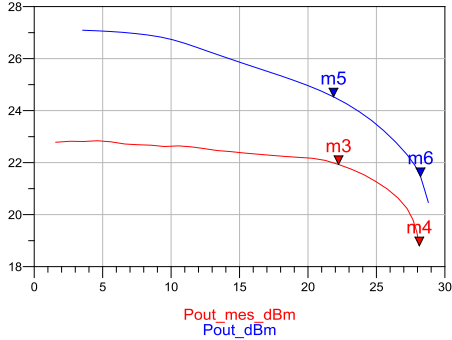
— Measurement
— Simulation



➤ Power Sweep Comparison

Cf Summary Table

GP vs OUTPUT POWER



m3
Pout_mes_dBm=22.236
plot_vs(PowerGain_mes_dB, Pout_mes_dBm)=21.929
idx_bloc=2.000

m4
Pout_mes_dBm=28.139
plot_vs(PowerGain_mes_dB, Pout_mes_dBm)=18.804
idx_bloc=2.000

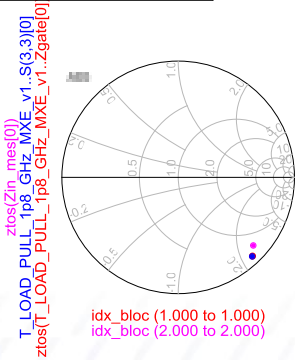
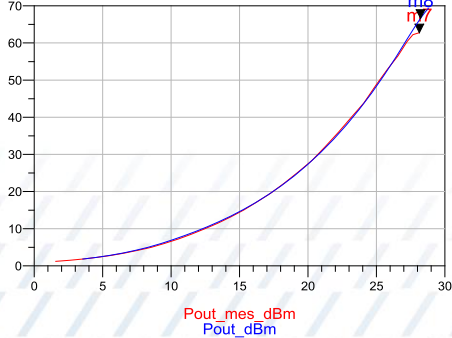
m7
Pout_mes_dBm=28.139
plot_vs(EFF_mes, Pout_mes_dBm)=62.745
idx_bloc=2.000

m5
Pout_dBm=21.866
plot_vs(T_LOAD_PULL_1p8_GHz_MXE_v1..Gp,T_LOAD_PULL_1p8_GHz_MXE_v1..Pout_dBm)=24.532
idx_bloc=1.000

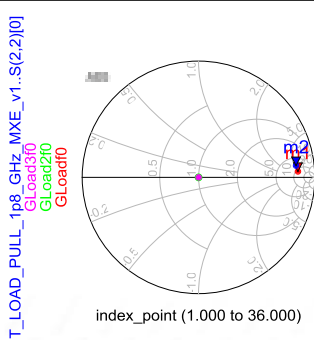
m6
Pout_dBm=28.223
plot_vs(T_LOAD_PULL_1p8_GHz_MXE_v1..Gp,T_LOAD_PULL_1p8_GHz_MXE_v1..Pout_dBm)=21.464
idx_bloc=1.000

m8
Pout_dBm=28.223
plot_vs(real(T_LOAD_PULL_1p8_GHz_MXE_v1..Eff), T_LOAD_PULL_1p8_GHz_MXE_v1..Pout_dBm)=66.63
idx_bloc=1.000

EFF vs INPUT POWER



idx_bloc	INRC
2.000	-6.640



idx_bloc	ONRC
2.000	-14.520

m1
index_point=36.000
GLoadf0=0.856 / 3.348
idx_bloc=2.000
impedance = 563.475 + j210.127

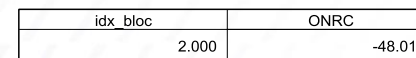
m2
indep(m2)=0
T_LOAD_PULL_1p8_GHz_MXE_v1..S(2,2)[0]=0.844 / 6.85
impedance = 395.076 + j276.079

Measurement Impedance : MXE & Simulation Impedance (MXE Simulated)

1.8 GHz at MXE – 28 V



Cf Summary Table



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LoadPull Performances

1.8 GHz – 28 V

— Measurement

— Simulation

MACOMTM

> Comparison Table

	MAX POWER PERFORMANCE			MAX EFF PERFORMANCE		
Output Impedance	235+j*17	230+j*11	-34.7	563+j*210	395+j*277	- 14.5 dB
Pout	20.2	20.2	0 dBm	27.9	27.8	-0.1 dB
Eff	57.2	56.2	- 1 pts	62.5	64.4	+ 1.9 pts
Gain	23.1	23.1	0 dB	22.3	24.7	+ 2.5 dB
Input Impedance	(cf next slide)			(cf next slide)		

LoadPull Performances

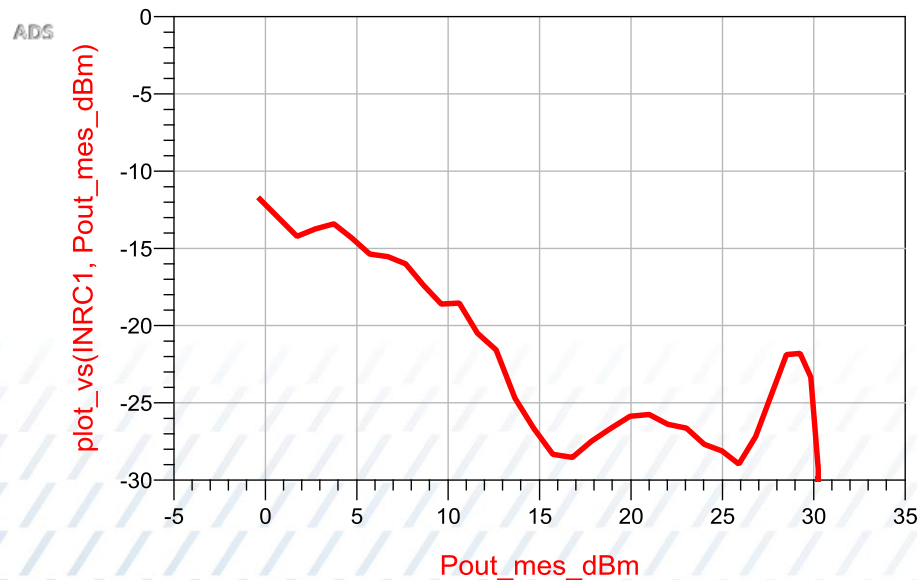
1.8 GHz – 28 V

➤ Input Normalized Reflexion Coefficient

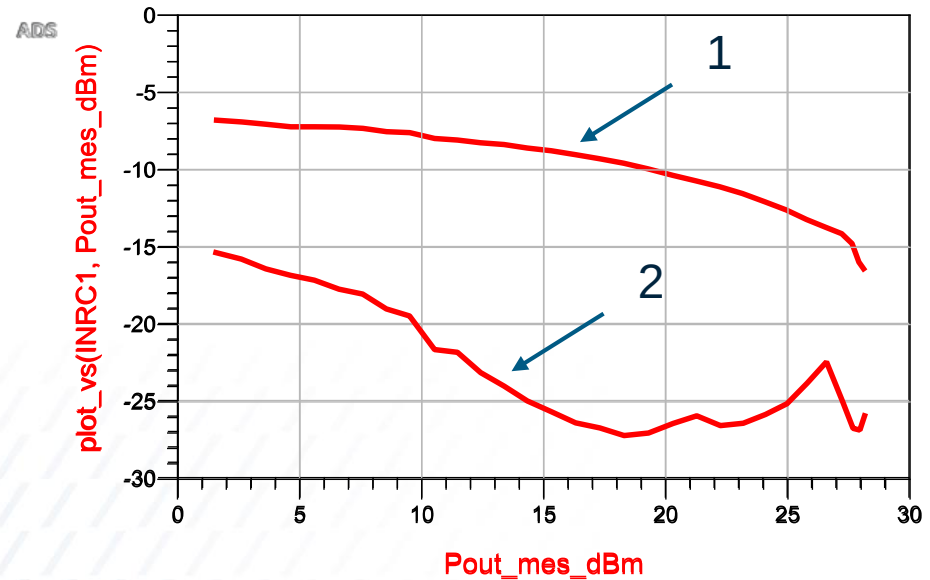
1 : Simulation at Z MXE Simulated

2 : Simulation at Z MXE Measured

@MXP



@MXE



- Pulsed IV
 - Thermal Validation
 - Id decreasing
 - Trap Validation
 - Transient versus Self Heating Ratio
- Pulsed S-Parameters
 - 28 V – 20 mA/mm
 - 50 V – 20 mA/mm
- Static S-Parameters
 - Class C & Class AB at 28 V
- **LoadPull Performances**
 - 1.8 GHz at MXP & MXE - 28 V
 - **3.8 GHz at MXP & MXE - 28 V**

LoadPull Performances

3.8 GHz at MXP – 28 V

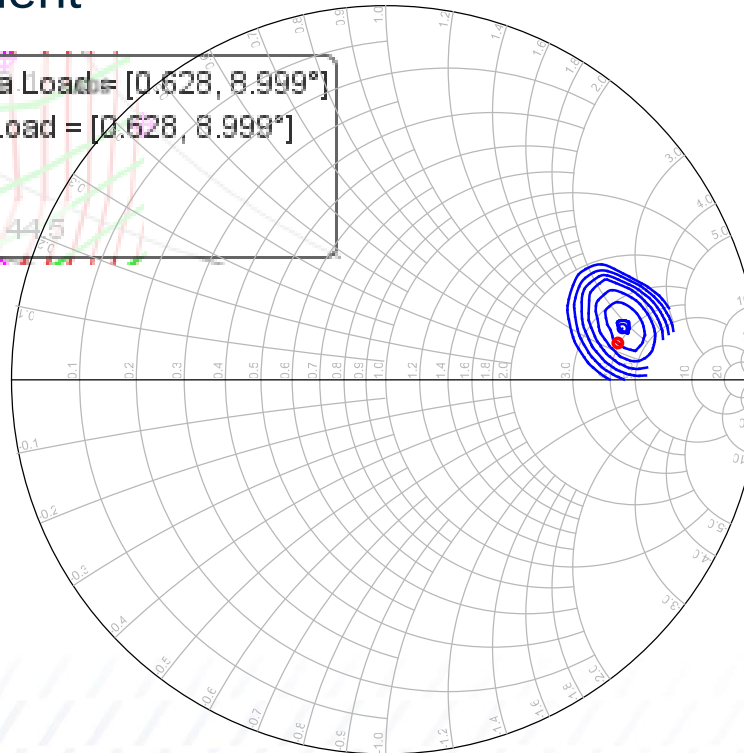
— Measurement
— Simulation

MACOM™

➤ From measurement

P_{out} = 29.87 dBm @ Gamma Load = [0.628, 8.999*]
PAE = 48.29 % @ Gamma Load = [0.628, 8.999*]
 Γ_{MXP} [0.628, 8.996*]
 196.8 + 63.79i ohms

ztos(Z_MXP)
 mPdel_vs_PAE_Rho
 Response3_Contours_XdB_Rho



Max &
 Contour levels, dBm:
 30.542
 30.400
 30.200
 30.000
 29.800
 29.600
 29.400

Zload_at mPdel vs PAE

188.889 + j88.889

PAE_at mPdel vs PAE

56.051

...dBm_at mPdel vs PAE

30.573

Gain_at mPdel vs PAE

15.726

Same for AMCAD

Eqn Z_MXP=196.8+j*63.79

Eqn Z_MXP_Simulation=stoz(mPdel_vs_PAE_Rho)

Eqn Coeff_Norm=db((Z_MXP-Z_MXP_Simulation)/(Z_MXP+conj(Z_MXP_Simulation)))

Coeff_Norm

-23.339

LoadPull Performances

3.8 GHz at MXP – 28 V

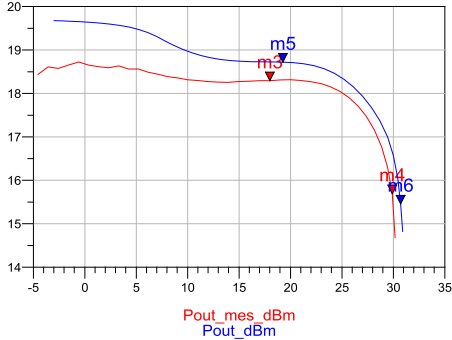
— Measurement
— Simulation



➤ Power Sweep Comparison

Cf Summary Table

GP vs OUTPUT POWER



m3
Pout_mes_dBm=17.981
plot_vs(PowerGain_mes_dB, Pout_mes_dBm)=18.291
idx_bloc=24.000

m4
Pout_mes_dBm=29.889
plot_vs(PowerGain_mes_dB, Pout_mes_dBm)=15.694
idx_bloc=24.000

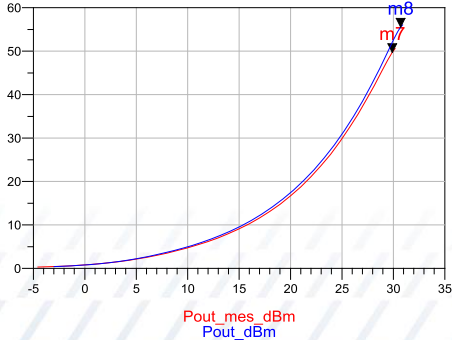
m7
Pout_mes_dBm=29.889
plot_vs(EFF_mes, Pout_mes_dBm)=49.708
idx_bloc=24.000

m5
Pout_dBm=19.256
plot_vs(T_LOAD_PULL_3p8_GHz_MXP_v1..Gp, T_LOAD_PULL_3p8_GHz_MXP_v1..Pout_dBm)=18.718
idx_bloc=1.000

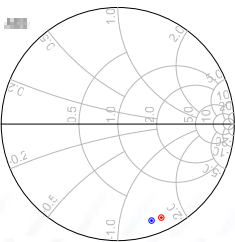
m6
Pout_dBm=30.704
plot_vs(T_LOAD_PULL_3p8_GHz_MXP_v1..Gp, T_LOAD_PULL_3p8_GHz_MXP_v1..Pout_dBm)=15.456
idx_bloc=1.000

m8
Pout_dBm=30.704
plot_vs(real(T_LOAD_PULL_3p8_GHz_MXP_v1..Eff), T_LOAD_PULL_3p8_GHz_MXP_v1..Pout_dBm)=55.55
idx_bloc=1.000

EFF vs INPUT POWER



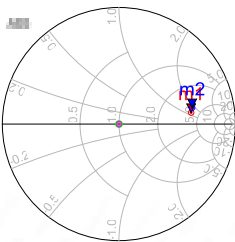
T_LOAD_PULL_3p8_GHz_MXP_v1..S(3,3)[0]
gamma_in_mes[0]



idx_bloc (24.000 to 24.000)

idx_bloc	INRC
24.000	-8.738

T_LOAD_PULL_3p8_GHz_MXP_v1..S(2,2)[0]
GLoad30
GLoad20
GLoad10



index_point (1.000 to 41.000)

m1
index_point=41.000
GLoad0=0.628 / 8.999
idx_bloc=24.000
impedance = 196.776 + j63.811

m2
indep(m2)=0
T_LOAD_PULL_3p8_GHz_MXP_v1..S(2,2)[0]=0.644 / 12.547
impedance = 185.672 + j88.652

idx_bloc	ONRC
24.000	-22.907

Measurement Impedance : MXP & Simulation Impedance (MXP Simulated)

LoadPull Performances

3.8 GHz at MXP – 28 V

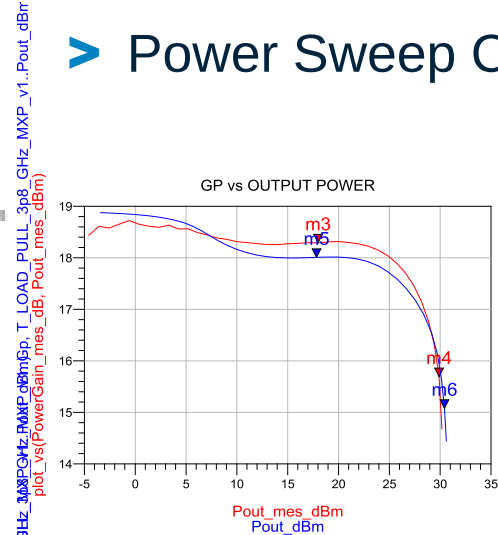
— Measurement
— Simulation



➤ Power Sweep Comparison

Cf Summary Table

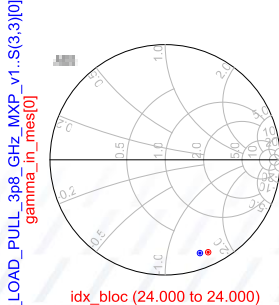
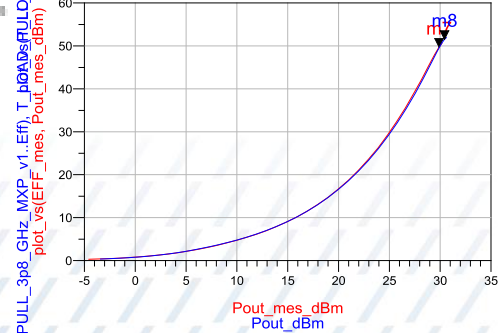
GP vs OUTPUT POWER



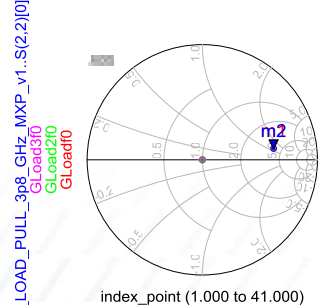
m3	Pout_mes_dBm=17.981 plot_vs(PowerGain_mes_dB, Pout_mes_dBm)=18.291 idx_bloc=24.000
m4	Pout_mes_dBm=29.889 plot_vs(PowerGain_mes_dB, Pout_mes_dBm)=15.694 idx_bloc=24.000
m7	Pout_mes_dBm=29.889 plot_vs(EFF_mes, Pout_mes_dBm)=49.708 idx_bloc=24.000

m5	Pout_dBm=17.837 plot_vs(T_LOAD_PULL_3p8_GHz_MXP_v1..Gp, T_LOAD_PULL_3p8_GHz_MXP_v1..Pout_dBm)=18.009 idx_bloc=1.000
m6	Pout_dBm=30.421 plot_vs(T_LOAD_PULL_3p8_GHz_MXP_v1..Gp, T_LOAD_PULL_3p8_GHz_MXP_v1..Pout_dBm)=15.079 idx_bloc=1.000
m8	Pout_dBm=30.421 plot_vs(real(T_LOAD_PULL_3p8_GHz_MXP_v1..Eff), T_LOAD_PULL_3p8_GHz_MXP_v1..Pout_dBm)=51.53 idx_bloc=1.000

EFF vs INPUT POWER



idx_bloc	INRC
24.000	-7.828



idx_bloc	ONRC
24.000	-45.878

m1	index_point=41.000 GLoadf0=0.628 / 8.999 idx_bloc=24.000 impedance = 196.776 + j63.811
----	---

m2	indep(m2)=0 T_LOAD_PULL_3p8_GHz_MXP_v1..S(2,2)[0]=0.626 / 8.876 impedance = 196.228 + j62.288
----	---

Measurement Impedance : MXP & Simulation Impedance (MXP Measured)

LoadPull Performances

3.8 GHz at MXE – 28 V

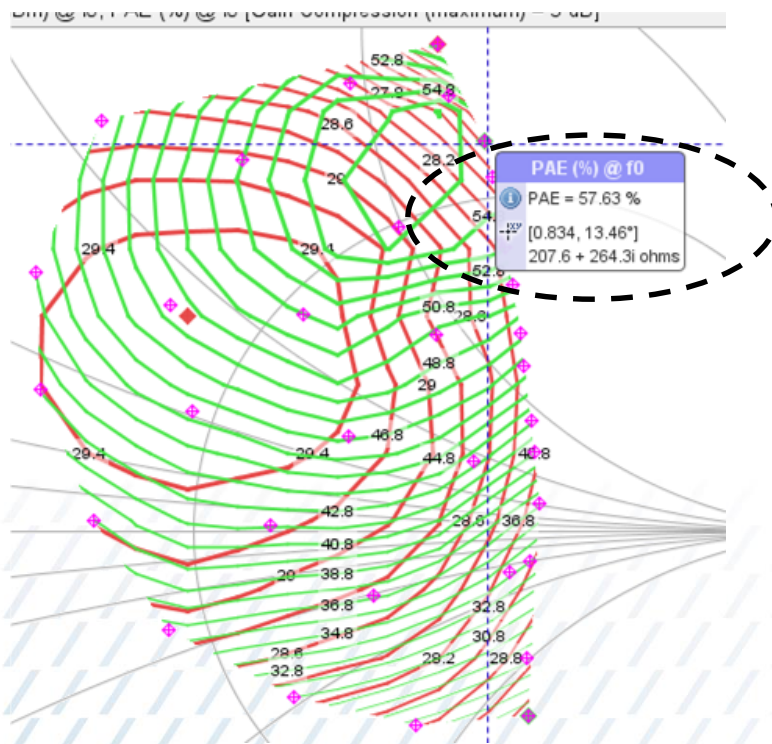
— Measurement

— Simulation

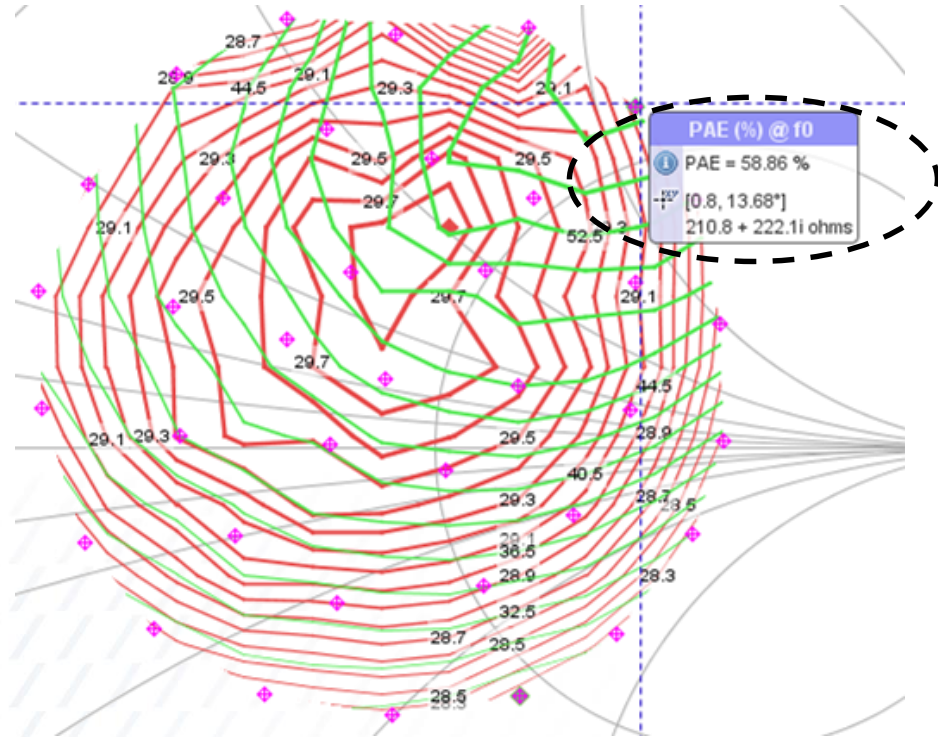
MACOMTM

➤ From measurement

- Maximum encircled and even better on “around MXP” sweep



MXE file



MXP file

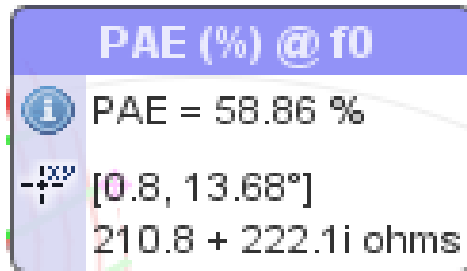
LoadPull Performances

3.8 GHz at MXE – 28 V

— Measurement
— Simulation

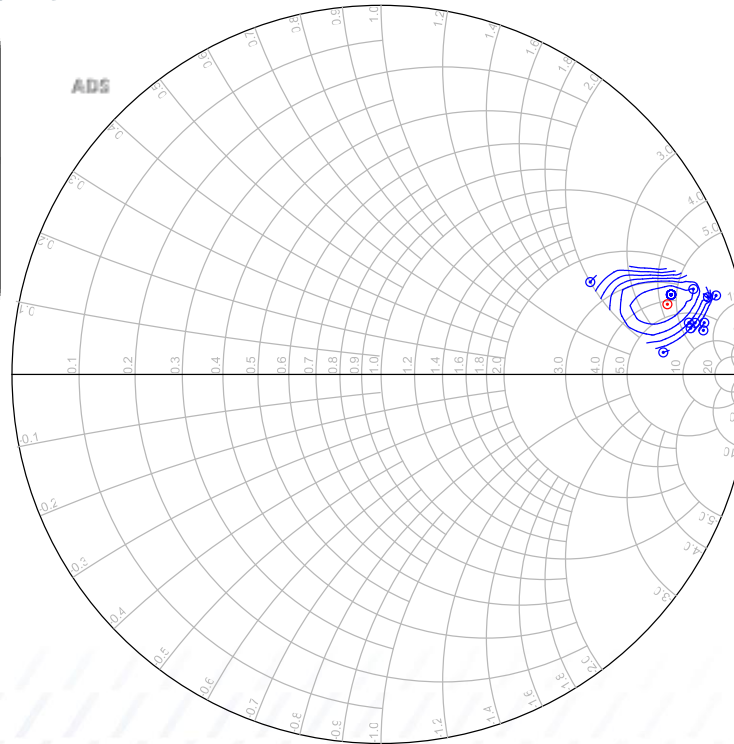
MACOM™

> From measurement



Same for AMCAD

z_{tos}(Z_MXE)
mPdel_vs_PAE
Response1_Contours



Maximum & Contour levels, %:

62.032
60.000
58.000
56.000
54.000
52.000
50.000
48.000

Zload_at_mPdel_vs_PAE	182.222 + j233.333
PAE_at_mPdel_vs_PAE	62.094
...el_dBm_at_mPdel_vs_PAE	28.776
Gain_at_mPdel_vs_PAE	16.832

Eqn Z_MXE=210+j*220

Eqn Z_MXE_Simulation=stoz(mPdel_vs_PAE_Rho)

Eqn Coeff_Norm=db((Z_MXE-Z_MXE_Simulation)/(Z_MXE+conj(Z_MXE_Simulation)))



LoadPull Performances

3.8 GHz at MXE – 28 V

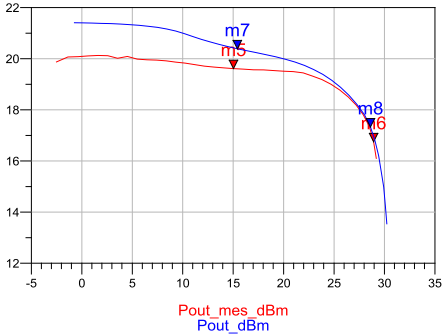
— Measurement
— Simulation



➤ Power Sweep Comparison

Cf Summary Table

GP vs OUTPUT POWER



m5
Pout_mes_dBm=15.035
plot_vs(PowerGain_mes_dB, Pout_mes_dBm)=19.615
idx_bloc=16.000

m6
Pout_mes_dBm=28.916
plot_vs(PowerGain_mes_dB, Pout_mes_dBm)=16.751
idx_bloc=16.000

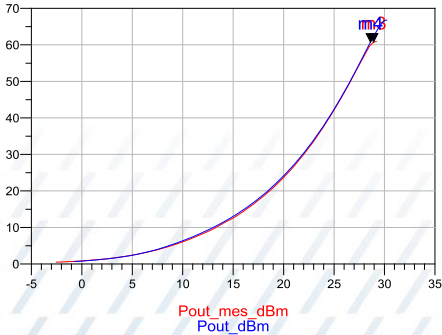
m3
Pout_mes_dBm=28.916
plot_vs(EFF_mes, Pout_mes_dBm)=60.771
idx_bloc=16.000

m7
Pout_dBm=15.420
plot_vs(T_LOAD_PULL_3p8_GHz_MXE_v1..Gp, T_LOAD_PULL_3p8_GHz_MXE_v1..Pout_dBm)=20.389
idx_bloc=1.000

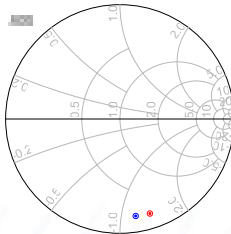
m8
Pout_dBm=28.610
plot_vs(T_LOAD_PULL_3p8_GHz_MXE_v1..Gp, T_LOAD_PULL_3p8_GHz_MXE_v1..Pout_dBm)=17.329
idx_bloc=1.000

m4
Pout_dBm=28.610
plot_vs(real(T_LOAD_PULL_3p8_GHz_MXE_v1..Eff), T_LOAD_PULL_3p8_GHz_MXE_v1..Pout_dBm)=60.736
idx_bloc=1.000

EFF vs INPUT POWER

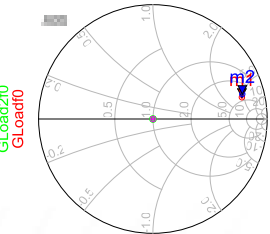


T_LOAD_PULL_3p8_GHz_MXE_v1..S(3,3)[0]
gamma_in_mes[0]



idx_bloc	INRC
16.000	-8.017

T_LOAD_PULL_3p8_GHz_MXE_v1..S(2,2)[0]
GLoad310
GLoad210
GLoad10



m1
index_point=41.000
GLoadf0=0.800 / 13.678
idx_bloc=16.000
impedance = 210.797 + j222.003

m2
indep(m2)=0
T_LOAD_PULL_3p8_GHz_MXE_v1..S(2,2)[0]=0.812 / 15.509
impedance = 180.414 + j229.360

idx_bloc	ONRC
16.000	-22.065

LoadPull Performances

3.8 GHz – 28 V

— Measurement

— Simulation

MACOMTM

> Comparison Table

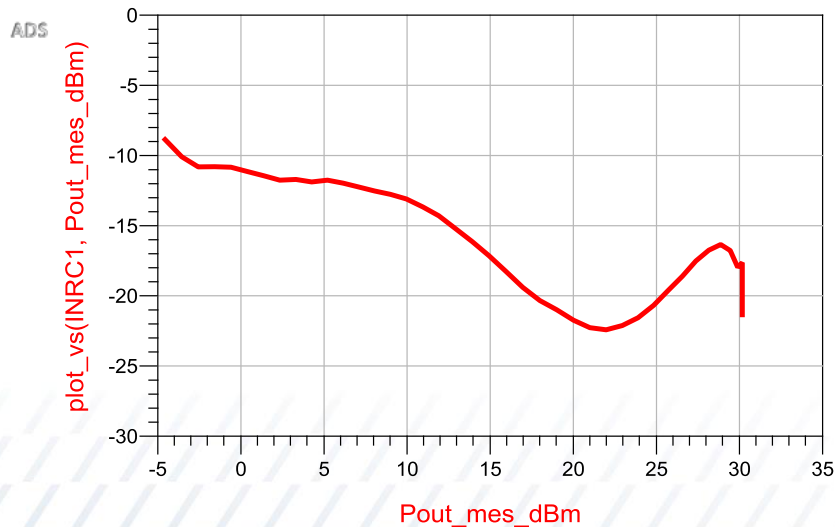
	MAX POWER PERFORMANCE			MAX EFF PERFORMANCE		
Output Impedance	196+j*63	188+j*88	- 23.4 dB	211+j*22	180+j*230	- 22 dB
Pout	29.9	30.7	+0.8 dB	28.9	28.6	- 0.3 dB
Eff	49.7	55.6	+ 5.9 pts	60.7	60.7	0 pts
Gain	18.3	18.7	+ 0.4 dB	19.6	20.4	+0.8 dB
Input Impedance	(cf next slide)			(cf next slide)		

LoadPull Performances

3.8 GHz – 28 V

➤ Input Normalized Reflexion Coefficient

@MXP



@MXE

